April 3, 2026

Good Friday Isaiah 49:16

NAWCWD Opportunities

Final Test Location & Times Options

- Tuesday April 21 7a-10a 325 EB
- Thursday Apr. 16 4p-9p 325 EB

Remediation

Case Studies
- Nitridation Process
- Liquid Nitrogen Freezing Process
- Classwork Submittal before 11:15a

Extra credit opportunities

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SEAP

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SMART
SCIENCE, MATHEMATICS, AND RESEARCH FOR TRANSFORMATION
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NREIP

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U.S. Chemical Safety and Hazard Investigation Board

Explosion, Molten Salt Eruption, and Fatal Injury at TS USA Liquid Nitriding Facility

Chattanooga, Tennessee | *Incident Date: May 30, 2024* | **No. 2024-01-I-TN**

Investigation Report

Published: June 2025



SAFETY ISSUES:

- Safety Management Systems
  - Procedures
  - Hazard Analyses
  - Training
- Learning from Past Incidents
- Corporate Engagement
  - Knowledge Management
  - Corporate Engagement





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EXECUTIVE SUMMARY

On May 30, 2024, at 8:58 a.m., an explosion occurred while employees were processing rollers with cavities at the TS USA facility in Chattanooga, Tennessee. The explosion caused an eruption of molten salts from the process, fatally injuring one TS USA employee and resulting in fires throughout the facility.

TS USA, a subsidiary of the French company HEF Groupe, treats metal parts in a liquid nitriding process, which involves a reaction to harden the surface of the metal. Leading up the incident, five rollers were submerged in molten salt baths used in the nitriding process. Each roller was a large cylinder that had an empty cavity with drain holes on the top and bottom. When the rollers were submerged in the baths, molten salt entered the cavities through the drain holes. Due to the small size of the drain holes, when the rollers were lifted from the salt bath to drain, the salt was unable to fully drain from one of the rollers and created a thick, solidified plug at the bottom of the roller. All five rollers were submerged in a hot water rinse bath to remove residual salt, but when the rollers were lifted for the water to drain, the salt plug prevented the water from draining from the plugged roller. Although the rollers were allowed to cool for over an hour, the plugged roller with the retained water remained at a temperature too hot to continuously touch.

When troubleshooting the plugged, hot roller, TS USA employees believed the internal cavity to be filled with solidified salt and that there was no water in the cavity. As such, a plan was developed to reintroduce all five rollers back into the salt bath to melt the solidified salt in the plugged roller so that salt could drain from the part. When introduced to the 800°F salt bath, the roller with the retained water began heating up. The water inside the roller began to boil, which rapidly increased the pressure inside the roller's cavity. The increasing pressure resulted in the roller failing and ejecting the bottom press fit end due to an overpressure, releasing steam and water into the molten salt. When the water contacted the high-temperature salt, it rapidly vaporized, resulting in a steam explosion. The steam displaced the hot molten salt, causing it to erupt from the bath and engulf a TS USA line operator, resulting in second- and third-degree chemical and thermal burns over 95 percent of his body. He was transported to a nearby hospital, where he died later that day.

The hot molten salts also ignited multiple fires throughout the facility. TS USA estimated that the eruption released 4,500 pounds of molten salt.

No off-site impacts were reported. The incident resulted in approximately $1.3 million in property damage. The facility was shut down for approximately eight months until it reopened in February 2025.

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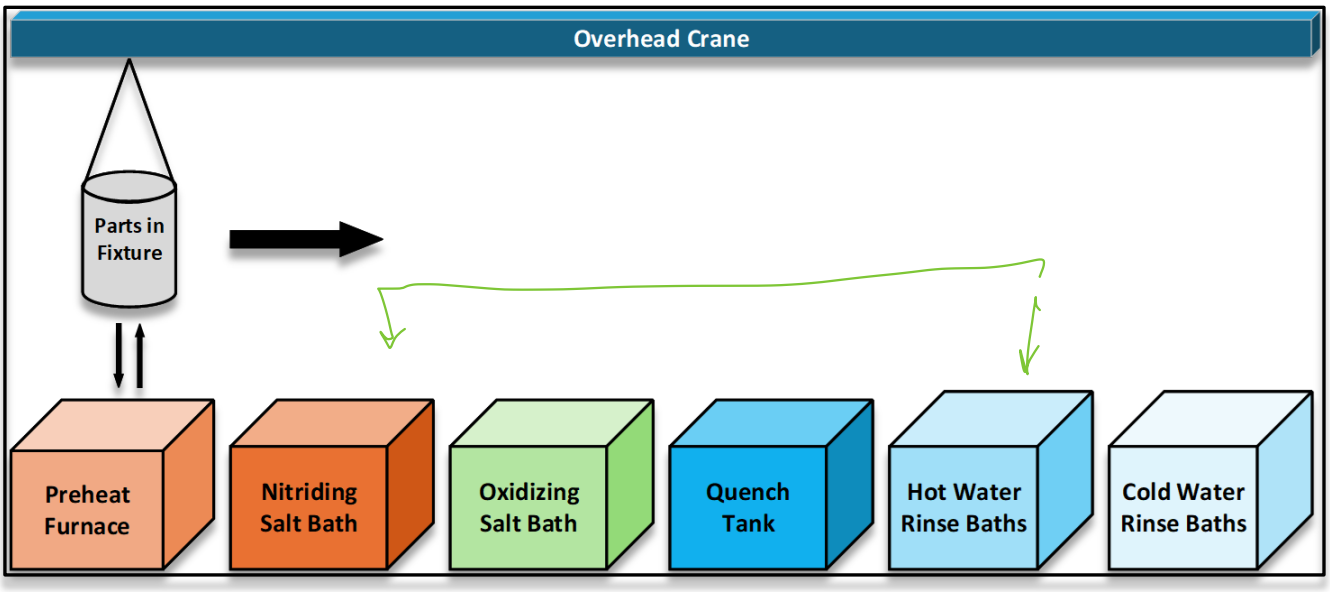


Figure 3. TS USA liquid nitriding line. (Credit: CSB)

After completing the nitriding process, metal parts continue to a surface finishing process, called polishing, and an optional oil impregnation step to increase corrosion resistance. Subsequently, the metal parts are reintroduced into the nitriding line, where the degreasing, water rinsing, preheat furnace, oxidizer bath, water quench, and water rinse bath steps are repeated to increase the parts' corrosion resistance. TS USA refers to this reintroduction process as the "re-oxidation process" [9].

1.3 Sodium Hydroxide and Sodium Nitrate

The incident at TS USA released a mixture of molten sodium hydroxide and sodium nitrate, called oxidizing salts, which are bright green in the molten state.[a] Sodium hydroxide, also known as lye or caustic soda, is a highly corrosive, strong base (high pH) that causes damage when in contact with human tissues, including the eyes, skin, and respiratory system. Sodium hydroxide is a white solid at room temperature and melts at 604°F.[b] Sodium hydroxide is non-combustible [10]. Sodium nitrate is a strong oxidizer that accelerates the burning of combustible materials. It is a white crystalline solid at room temperature and melts at 584°F [11].

1.4 Metal Parts Processed on May 30, 2024

During the incident, TS USA was processing cylindrical metal parts, called rollers. The rollers were constructed of 316 stainless steel and were approximately 7.4 feet (2.3 meters) long. The rollers had an approximately 5-inch (120-millimeter) outer diameter and contained a 4-inch (100-millimeter) cavity through the largest diameter portion, as shown in **Figure 4**. The cavity was enclosed by press fit inserts at each end that contained five blind holes, which are holes that are not drilled entirely through the material [12, p. 95]. The press fit insert also had

[a] At the moment of the release, the molten oxidizing salt mixture was approximately 800˚F.
[b] When cooled below 604˚F, molten sodium hydroxide returns to a solid state.

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an approximately 0.2-inch (5-millimeter) diameter hole on either end of the roller that penetrated to the cavity interior, as shown in **Figure 5**. There were no other openings to the interior cavity.



Figure 4. Metal roller containing a cavity that TS USA processed during the incident. The annotated area is approximate. (Credit: CSB)

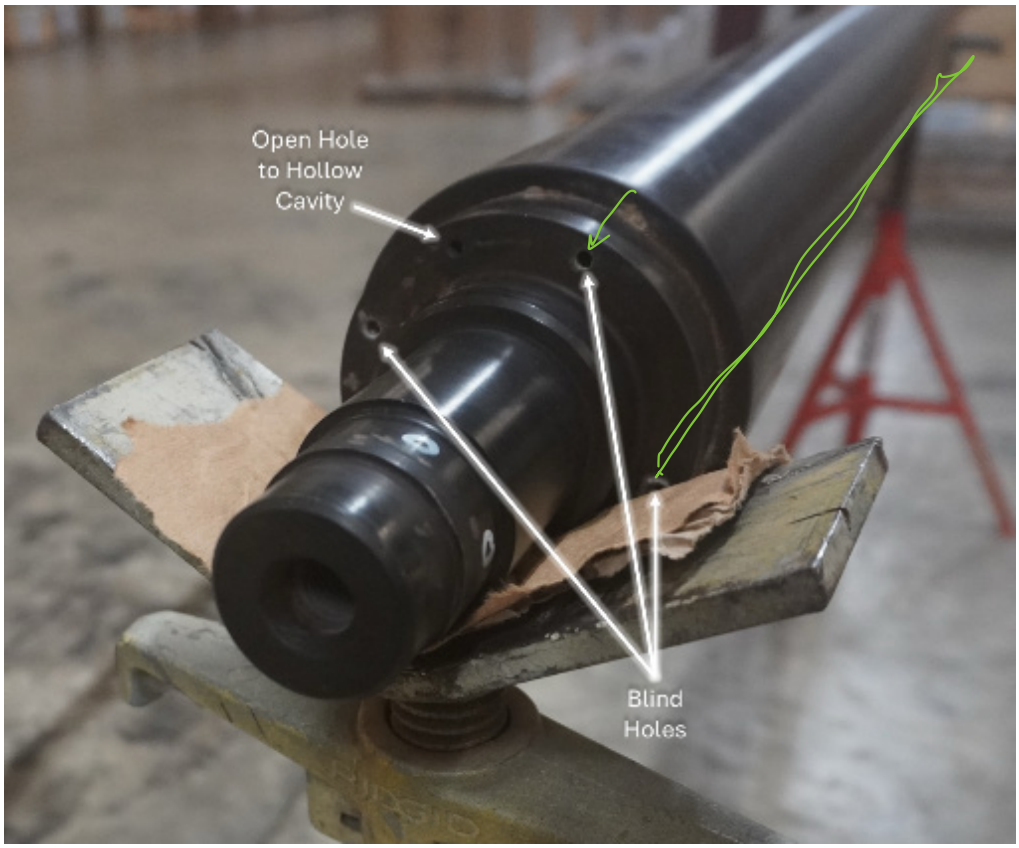


Figure 5. Open hole to the interior cavity on rollers. (Credit: CSB)

1.5 Regulatory Coverage

The Tennessee Occupational Safety and Health Administration (TOSHA) oversees an Occupational Safety and Health Administration (OSHA) approved state program and is responsible for Process Safety Management (PSM) compliance in Tennessee. The TS USA Chattanooga facility did not process or store any materials

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the rapid vaporization of water, also known as a vapor or steam explosion.[a] The rapid expansion of the steam displaced the molten salt in the oxidizer bath, resulting in molten salt erupting from the vessel (**Figure 14.D**).

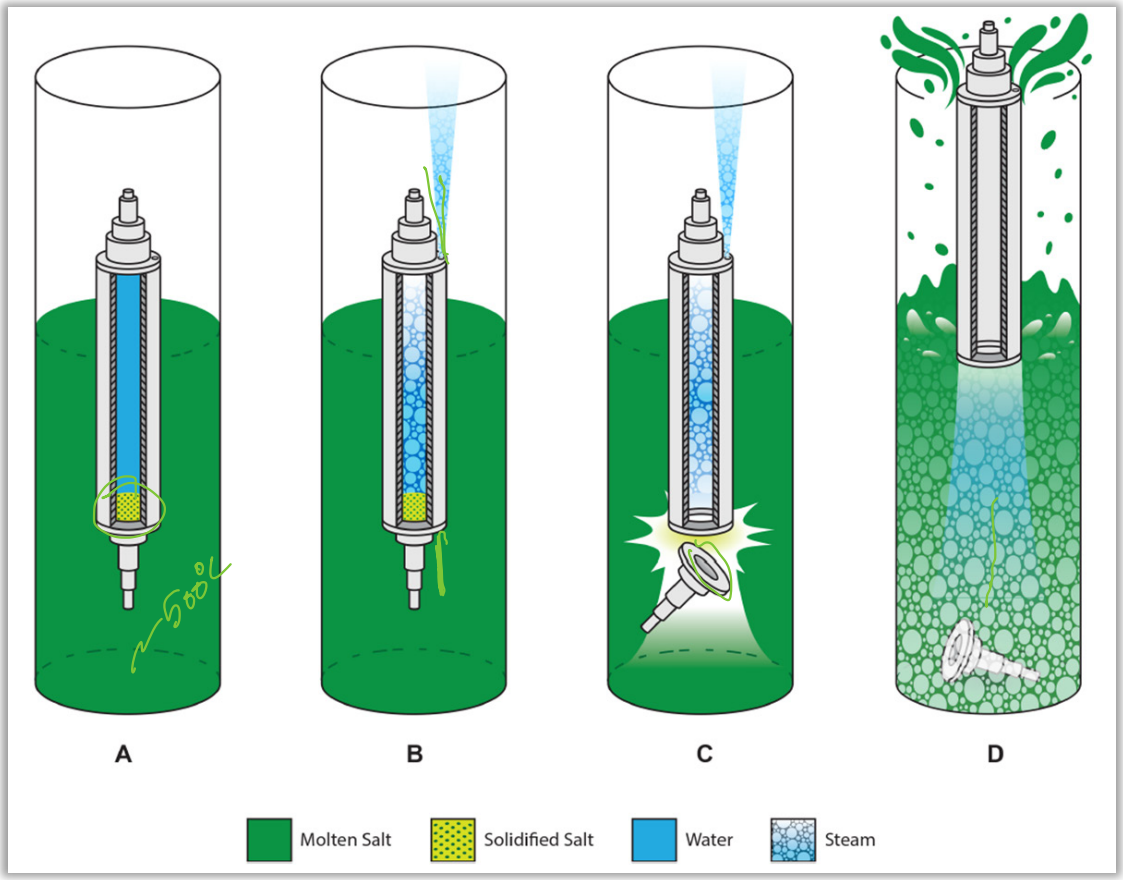


Figure 14. Sequence of events of the overpressure: A) The roller is placed in the molten salt bath. B) Water boiled and vented from the roller. C) Overpressure caused the press fit end to dislodge. D) Rapid expansion of steam and molten salt eruption. (Credit: CSB)

The CSB concludes that the retained water boiled inside the roller's cavity and rapidly expanded as steam. The increased pressure forcefully ejected the roller's press fit end, the salt plug, and the water into the molten salt bath. The water rapidly boiled, creating a violent steam explosion that drove an uncontrolled eruption of molten salt from the vessel.

[a] A vapor explosion can occur on contact between two liquids of differing temperatures if the temperature of the hotter liquid is above the boiling point of the cooler, and the explosion is due to extremely rapid vapor generation (phase transition) of the cooler liquid [53, p. 1392].

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4.1.2 HAZARD ANALYSES

The CCPS's *Guidelines for Risk Based Process Safety* element *Hazard Identification and Risk Analysis* encompasses all activities involved in identifying hazards and evaluating risk at a facility throughout the facility's life cycle, to ensure that risks to workers, the public, and the environment are consistently controlled. The analysis should assess and ask:

- Hazard – What can go wrong?
- Consequences – How bad could it be?
- Likelihood – How often might it happen [25, p. 210]?

The understanding of risk developed from these exercises helps form the basis for establishing most of the other process safety management activities undertaken by the facility [25, p. 211]. A process hazard analysis (PHA) is typically performed to meet industry guidance for hazard identification and risk analysis [25, p. 210]. Volume 20 of the ASM International Handbook, *Materials Selection and Design*, provides guidance for risk analysis through multiple methodologies, including Failure Modes and Effects Analysis (FMEA) and Fault Tree Analysis [29, pp. 117-125]. PHAs are required by the NFPA for all furnaces covered by NFPA 86, *Standard for Ovens and Furnaces* [15].[a]

The hierarchy of controls should be considered when assessing hazard controls and determining safeguard effectiveness. The hierarchy of controls (**Figure 15**) is a method of describing the effectiveness or preference of safeguard implementation.

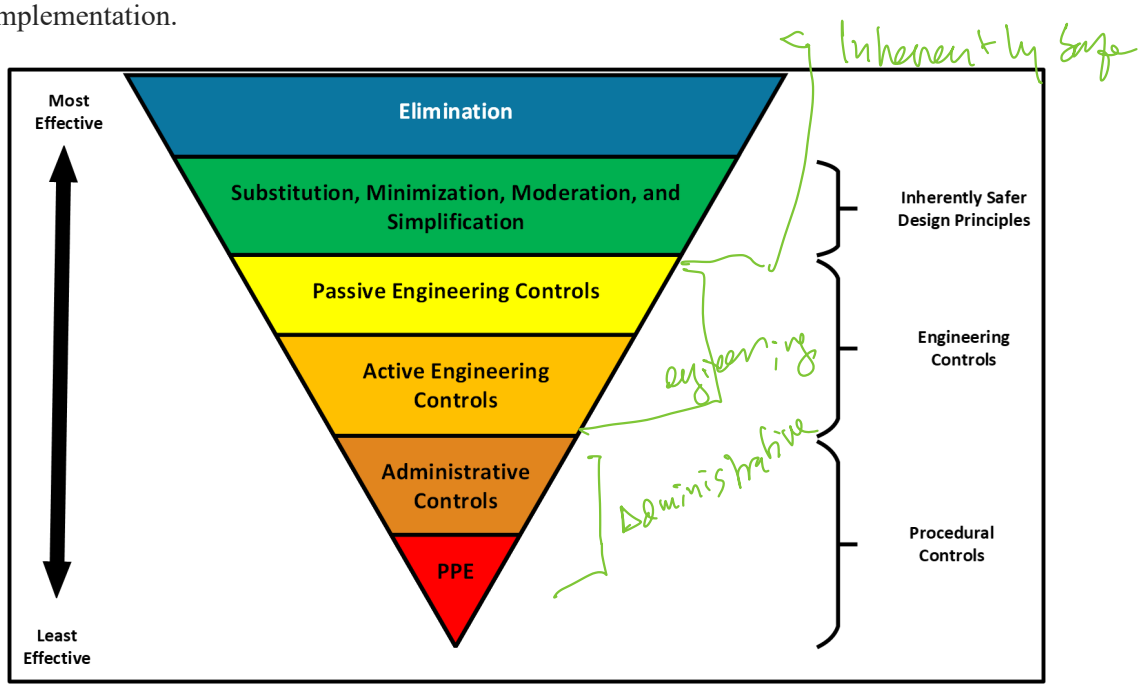


Figure 15. Hierarchy of controls. (Credit: CSB, modified from CCPS [30] and NIOSH [31])

[a] The NFPA defines Class C furnaces as "An oven or furnaces that has a potential hazard due to a flammable or other special atmosphere being used for treatment of material in process" and specifically identifies molten salt baths as this type of furnace. Annex N provides additional details for the frequency and contents of process hazard analyses to be performed as required by NFPA 86, *Standard for Ovens and Furnaces* [15].

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In *Guidelines for Risk Based Process Safety* and *Guidelines for Investigating Process Safety Incidents*, the CCPS recommends that companies broadly communicate investigative findings internally and externally [25, pp. 561-562, 40, p. 345]. Communicating investigation results within the company where the incident occurred allows all involved parties to be aware of the causes and lessons learned; additionally, internal communication allows a company to track corrective actions to determine whether they led to improvements [25, p. 562]. Sharing the results between different facilities within the same company allows the causes and successful corrective actions to be communicated to facilities that perform similar or identical processes; external communication helps prevent similar incidents at different facilities [25, p. 562].

The CSB identified three incidents that occurred at TS facilities. TS USA did not effectively learn from these incidents, however, because it did not investigate incidents at its facilities and did not apply any lessons from previous incidents. None of these three incidents were effectively communicated to other facilities within HEF Groupe.

KEY LESSON

Companies should have an incident investigation program that generates formal reports, performs causal analysis, and reviews corrective actions. The findings of the investigations should be communicated, including translation, throughout the site and to other facilities within the company.

4.2.1 2018 Event in Mexico

On January 25, 2018, an explosion occurred at a TS ETSA facility in Aguascalientes, Mexico, that injured three workers and caused serious damage at the facility. TS ETSA was hired to treat an 8-foot (2.3-meter) long cylindrical roller with an 18-inch (450-millimeter) outer diameter, and an internal coil that spiraled through the center (**Figure 20**). This roller was designed so that a fluid could cool the roller while it was operating.

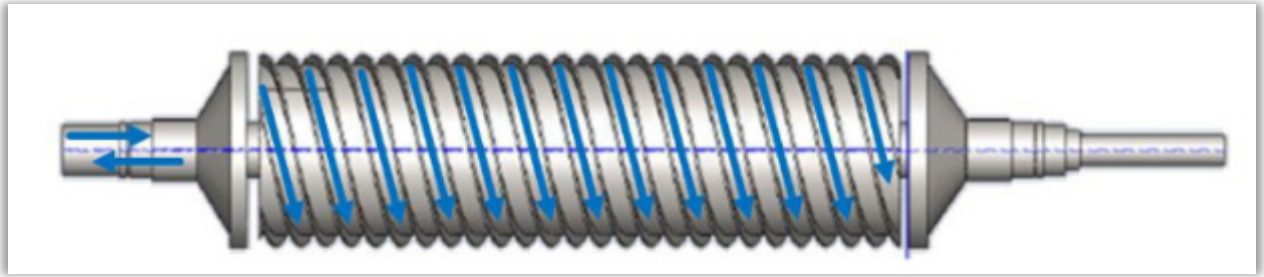


Figure 20. A diagram of the roller involved in the Mexico explosion. (Credit: TS ETSA)

On the day of the incident, a cap was placed over the end where the fluid could enter and leave; however, TS ETSA's investigation found that the cap was not correctly placed over the opening. TS ETSA positioned the roller so that the capped end was facing downward when it was hooked up to the crane. The roller was then processed through the degreasing, cleaning, and rinse baths; water likely entered the roller during the rinsing step. The roller was then sent to the preheating step. TS ETSA measured that the roller reached a temperature of approximately 400°F during preheating.

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After preheating, the roller was introduced to the nitriding salt bath, which was approximately 1,040°F. Shortly after the introduction to the nitriding salt bath, the roller exploded. The high temperature caused the water to expand and rapidly increased the pressure until the cap was no longer able to contain it. The cap was forcefully ejected from the roller, relieving the built-up pressure in an overpressure explosion, similar to the conditions discussed in **Section 3.2**.

The explosion ejected molten salt from the bath and launched the roller across the facility, where it impacted several areas, destroyed the nitriding salt bath's furnace, cracked a wall at the facility, deformed the crane's support structure, destroyed the crane's hoist, and tore a hole in the roof. In addition to the damages caused by the launched roller, the ejected salts damaged the electrical control boards at the facility and sprayed onto the building's roof and onto the roofs of neighboring buildings. The ejected salt contacted three employees, causing first- and second-degree burns and eye damage.



Figure 21. The visual aid created by TS ETSA for part inspections. (Credit: TS ETSA, translated by the CSB)

TS ETSA investigated the incident and developed a formal report detailing the incident, causal factors, and corrective actions. TS ETSA determined that the cause of the incident was that a part with an accumulation hazard was allowed to be processed in the liquid nitriding line. As part of TS ETSA's corrective actions, a pre-processing inspection is now required at the facility before parts are allowed to be processed, and a visual aid was created to help employees with the inspection (**Figure 21**). The visual aid showed operators specific items to look for during the inspection, such as hollow cavities, welds, or sealed parts.

The investigation report for the 2018 Mexico explosion only existed in Spanish and required translation by the CSB. A significantly less detailed report was created, translated to English, and presented to HEF Groupe, however. The English version of the report omitted many details and is only five pages long, while the Spanish version is 44 pages long. Additionally, no translated version of the visual aid existed, and in order to read it, it required translation by the CSB.

The TS USA plant managers told the CSB that they were not familiar with the incident at the TS ETSA facility in Mexico. One plant manager said that he "learned [about the Mexico explosion] by accident." Another said,

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RECOMMENDATIONS

To TS USA

2024-01-I-TN-R1

Implement physical, protective barriers around the molten salt baths that isolate employees from hazardous releases at all locations that perform liquid nitriding.

2024-01-I-TN-R2

Develop a safety management system that incorporates industry guidance and includes, but is not limited to:

a. A hazard analysis program for assessing the nitriding process. The program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety*, the National Fire Protection Association's (NFPA) *Standard for Ovens and Furnaces*, and ASM International's Handbook. The program shall apply to new and existing parts, assess parts for accumulation hazards and sealed cavities, and include non-routine tasks such as reprocessing unsatisfactory parts.

b. Written operating procedures for the nitriding process. The procedures shall be based on the information gathered from the hazard analysis program. The procedures shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety* and the CCPS's *Guidelines for Writing Effective Operating and Maintenance Procedures*.

c. A training program, including written materials, for the employees involved in the nitriding process. This program shall be based on the nitriding facility's operating procedures and other relevant information from the hazard analysis program. The program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety*.

d. An incident investigation program. This program shall include requirements for performing causal analysis, producing written reports, and communicating findings and corrective actions throughout the entire TS USA organization. The program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety* and the CCPS's *Guidelines for Investigating Process Safety Incidents*.

2024-01-I-TN-R3

For each TS USA facility, establish a position with specific professional expertise and experience in safety management systems, such as risk-based process safety. This position shall be responsible for TS USA's safety management system, ensuring that HEF Groupe's safety information is incorporated at the site level, and implementing regulatory and industry safety guidance.

To HEF Groupe

2024-01-I-TN-R4

Include physical, protective barriers as part of the standard design for liquid nitriding processes. These protective barriers shall be intended to isolate employees from molten salt releases.

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2024-01-I-TN-R5

Develop a safety management system that incorporates industry guidance and includes, but is not limited to:

a. A hazard analysis program for assessing the nitriding process. The program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety*, the National Fire Protection Association's (NFPA) *Standard for Ovens and Furnaces*, and ASM International's Handbook. The program shall apply to new and existing parts, assess parts for accumulation hazards and sealed cavities, and include non-routine tasks such as reprocessing unsatisfactory parts.

b. Written operating procedures for the nitriding process. The procedures shall be based on the information gathered from the hazard analysis program. The procedures shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety* and the CCPS's *Guidelines for Writing Effective Operating and Maintenance Procedures*.

c. A training program, including written materials, for the employees involved in the nitriding process. This program shall be based on the nitriding facility's operating procedures and other relevant information from the hazard analysis program. The program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety*.

d. An incident investigation program. This program shall include requirements for performing causal analysis, producing written reports, and communicating findings and corrective actions throughout the entire HEF Groupe organization. The program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Risk-Based Process Safety* and the CCPS's *Guidelines for Investigating Process Safety Incidents*.

2024-01-I-TN-R6

Develop and implement an effective and comprehensive Knowledge Management program for sharing knowledge throughout the HEF Groupe organization. Knowledge shall include all information from audits, hazard analyses, and incident investigations, including causal analyses and corrective actions recommended and taken. This program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Process Safety Knowledge Management*.

2024-01-I-TN-R7

Develop and implement a comprehensive and effective Corporate Governance program. This program shall include regular audits of subordinate facilities throughout the organization, with tracking and accountability for implementation of all recommendations and corrective actions identified in the audits. Facility adherence to the safety management system recommended above shall be evaluated during the audits. The program shall require documentation of audit findings, prompt responses to deficiencies, development of corrective actions, and implementation of the corrective actions throughout the organization. This program shall incorporate industry guidance, such as the Center for Chemical Process Safety's (CCPS) *Guidelines for Implementing Process Safety Management*.

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5 CONCLUSIONS

5.1 FINDINGS

Technical Analysis

1. The roller's design allowed for the accumulation of molten salt in the roller's interior cavity, which solidified into a salt plug.

2. TS USA did not recognize the accumulation hazard presented by the cavity, which, when the molten salt solidified, blocked the drain hole in the bottom of the roller.

3. The rollers were not suitable for processing through the liquid nitriding process. The drain holes were not large enough to ensure the rollers fully drained the molten salts, resulting in the salt plug forming at the bottom of the roller's interior.

4. The retained water boiled inside the roller's cavity and rapidly expanded as steam. The increased pressure forcefully ejected the roller's press fit end, the salt plug, and the water into the molten salt bath. The water rapidly boiled, creating a violent steam explosion that drove an uncontrolled eruption of molten salt from the vessel.

Safety Management Systems

1. The operating manual used by TS USA employees did not provide sufficient information to ensure that employees were aware of the potential hazards in the nitriding operation.

2. TS USA and HEF Groupe relied on the oxidizing salt mixture safety data sheet (SDS) for safety instructions in its operating manual, but the SDS does not provide sufficient information on hazards for materials in the process, which should be incorporated in the operating manual.

3. TS USA lacked procedures for reprocessing the rollers on the day of the incident. Had TS USA had procedures for reprocessing operations, it could have ensured that risks were identified and that safeguards were in place to mitigate the hazards presented by the operation.

4. TS USA deviated from its operating manual and HEF Groupe's requirements by reintroducing parts to the oxidizer bath and skipping the required preheating step.

5. TS USA did not perform any safety-based risk assessments on the liquid nitriding process. Had TS USA applied the methodology used for its quality program for process hazard analyses, it could have identified hazards, such as the risk of accumulations in parts, reactions due to foreign material in the molten salt baths, and the potential for upset conditions, such as overpressure events or explosions.

6. HEF Groupe assessed the potential hazards presented by an explosion and release of molten salts. However, the risk analyses did not incorporate inherently safer design principles or engineering controls. Had HEF Groupe implemented additional controls, it could have prevented the release of the molten salt and the explosion.

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7. The risk analyses were not communicated to the TS USA facilities. Had the risk analyses been communicated, the hazard of water introduction into the bath and the requirement for all parts to be fully dried could have been followed, and the cause of the explosion eliminated.

8. The additional passive engineering controls in place at the Chattanooga facility were insufficient to protect the line operator due to the severity of the eruption of molten salt, which was more significant than the installed barrier was designed to withstand and protect against.

9. TS USA viewed the rollers as safe to process once the sealed cavity was opened and did not recognize the accumulation hazard in the newly opened cavity. Had TS USA recognized the risk of accumulations in the cavity, it could have requested that the customer take additional actions, such as removing the press fit ends, to eliminate the hazard.

10. The review process for new parts used by TS USA and HEF USA did not adequately assess the potential accumulation hazards or hazards due to modifications to new parts. The review process did not include corporate and industry guidance, such as the prohibition of processing sealed parts or parts that had not been dried, which could have mitigated or prevented the incident. Had TS USA's review included the guidance to prohibit the processing of parts that could accumulate materials, it could have decided that the rollers were not suitable for the nitriding operation.

11. Adding drainage holes to parts with sealed cavities directly contributed to the fatal May 2024 eruption at the Chattanooga facility. Had the practice of adding through-holes been thoroughly assessed before it was implemented, the new hazards, such as the accumulation in the roller cavity, could have been identified and mitigated.

12. TS USA did not perform an additional hazard analysis following the initial trial run and failed to recognize the accumulation hazard in the rollers. Had TS USA performed a hazard analysis, TS USA could have taken additional steps, such as removing the press fit ends, improving drainage, eliminating the accumulation hazard, and providing access to the internal cavity.

13. TS USA did not perform a hazard analysis on the non-routine task for reworking the rollers. Had TS USA performed a hazard analysis, it could have identified that the rollers were filled with water, and that reintroducing the rollers to the oxidizer bath could be a hazard.

14. TS USA's training program did not provide information or guidance on the process hazards. Had TS USA provided training on the hazards of the process and the potential for explosions, the risk of water accumulation and the potential for an explosion could have been identified, the operation stopped, and the incident prevented.

15. TS USA did not have a training program on how to identify hazards in the process or when evaluating new parts. Had TS USA developed and trained its reviewers on potential hazards when evaluating new parts, the reviewers could have identified the accumulation risks present in the rollers.

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Learning from Past Incidents

1. The 2018 explosion in Mexico, the explosion's causal factors, and the corrective actions taken were not communicated from HEF Groupe to the other facilities. Had the corrective actions from the Mexico incident been communicated, TS USA could have identified the accumulation hazard and rejected the rollers during the pre-processing safety review, which could have prevented the 2024 incident.

2. No investigation report or corrective actions were distributed to TS USA or other HEF Groupe facilities following the 2020 explosion in France, and the CSB cannot confirm whether a formal investigation was performed or a written investigation report produced after the incident. The lack of documentation prevented the communication of the incident, its causal factors, corrective actions (if any), and lessons learned to other facilities within HEF Groupe.

3. The 2023 Chattanooga incident and its causal factors were not formally or robustly investigated, and any corrective actions taken were not communicated to other TS USA facilities. Had the 2023 Chattanooga incident been formally investigated and findings communicated, the safety personnel at HEF Groupe could have investigated the incident and communicated HEF Groupe's policy not to process parts that could accumulate materials. Had TS USA been aware of the safety concerns associated with processing parts that accumulate materials, the fatal 2024 incident could have been prevented.

4. TS USA, HEF Groupe, and other TS facilities have repeatedly had reason to understand that parts with accumulation hazards and sealed cavities are not suitable for the liquid nitriding process. All three incidents discussed above—2018 in Mexico, 2020 in France, and 2023 in Chattanooga—show that these parts are susceptible to trapping materials in their cavities, which leads to overpressure explosions when the parts are introduced to the salt baths, making them unsafe to process.

5. TS USA's and HEF Groupe's failure to learn from these previous incidents led to missed opportunities to identify the conditions that contributed to the 2024 explosion.

Corporate Engagement

1. HEF Groupe did not communicate the risk assessments performed by HEF Groupe on the nitriding process. Had HEF Groupe communicated the risk assessments and the controls for the identified hazards, TS USA could have ensured that the necessary safeguards were in place when processing parts through the nitriding line, and the 2024 incident likely could have been prevented.

2. HEF Groupe did not sufficiently share critical process safety knowledge with its subsidiaries, particularly by not communicating and explaining the hazards of accumulations in parts with cavities. Had HEF Groupe provided TS USA with the knowledge to assess the potential hazards presented by the rollers with the information in the risk analysis, TS USA could have recognized the accumulation hazard.

3. HEF Groupe did not manage safety knowledge throughout the company. HEF Groupe did not ensure that its subsidiary companies were provided with all the safety information that had been developed or that safety knowledge had been transferred and managed at its facilities. Had HEF Groupe managed its safety knowledge appropriately and ensured that safety information was incorporated into the operations

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of each of HEF Groupe's facilities, TS USA could have been aware of the hazards, mitigated the risk of fatally injuring an employee, and likely prevented the fatal 2024 incident in Chattanooga.

4. HEF Groupe did not provide the resources necessary to ensure that safety requirements and management systems were in place at its facilities. Had HEF Groupe and TS USA leadership assigned safety resources and responsibilities to dedicated roles, the safeguards and barriers identified to mitigate explosions could have been maintained and their implementation verified at the facilities, including the Chattanooga facility.

5.2 CAUSE

The CSB determined that the cause of the incident was the introduction of water contained in the roller cavity to the 800°F oxidizing salt bath. The hot salt caused the water to expand and boil in the cavity of a roller, which resulted in an overpressure, a steam explosion, and a molten salt eruption.

Contributing to the incident was TS USA's lack of awareness of the accumulation hazards associated with parts containing cavities. These cavities presented accumulation hazards when processed in the nitriding line. Also contributing to the incident were TS USA's and HEF Groupe's insufficient process safety management systems, which did not include adequate procedures, training, hazard analyses, and incident investigations. Also contributing to the incident was HEF Groupe's ineffective corporate governance and safety knowledge management, which did not ensure that critical safety information was communicated and accessible to all TS USA facilities, including details of prior safety incidents and the results of those investigations.

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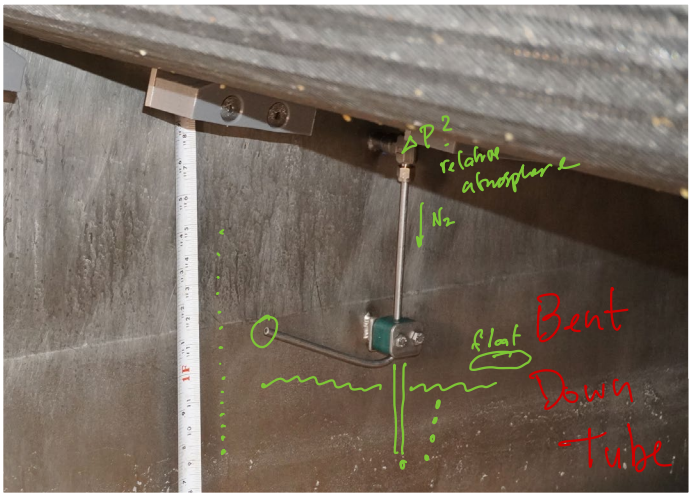
U.S. Chemical Safety and Hazard Investigation Board

Fatal Liquid Nitrogen Release at Foundation Food Group

Gainesville, GA | *Incident Date: January 28, 2021* | **No. 2021-03-I-GA**

Investigation Report

Published: December 2023



SAFETY ISSUES:

- Single Point of Failure
- Atmospheric Monitoring and Alarm Systems
- Emergency Preparedness
- Process Safety Management System
- Product Stewardship



EXECUTIVE SUMMARY

On Thursday, January 28, 2021, between approximately 8:45 and 10:15 a.m., liquid nitrogen overflowed from an immersion freezer located inside the Plant 4 building at the Foundation Food Group (FFG) facility in Gainesville, Georgia. The release began while maintenance workers were troubleshooting operational issues with the freezer, which was designed and owned by Messer LLC (Messer) and leased to FFG. Once released, the liquid nitrogen quickly vaporized, expanded, and accumulated inside a partially enclosed lower-level room that lacked mechanical ventilation. The two maintenance workers who were troubleshooting the freezer at the time of the release were fatally injured by asphyxiation due to the vaporized liquid nitrogen.

The uncontrolled liquid nitrogen release continued, and the two deceased maintenance workers went undetected for 30 to 60 minutes, until another worker went looking for them and saw a four- to five-foot-high vapor cloud filling the room. This worker reported the incident to management, which then initiated an evacuation. During the ensuing building-wide evacuation, at least 14 other FFG employees, including members of management, responded to the incident by either investigating the freezer room or attempting to rescue their coworkers. As a result, four additional FFG employees were fatally injured by asphyxiation, and three other FFG employees and a firefighter were seriously injured presenting asphyxiation symptoms.

Court filings show that after the incident FFG sued its insurance company for damages of roughly $1.7 million. Messer reported business and property losses of roughly $245,000.

SAFETY ISSUES

The CSB's investigation identified the safety issues below.

- **Single Point of Failure.** The immersion freezer design included a device called a bubbler tube, which was used to measure the liquid nitrogen level inside the freezer. The bubbler tube was likely bent during maintenance activity, rendering it unable to measure and control the freezer's liquid nitrogen level. As a result, liquid nitrogen overflowed from the freezer and filled the room with vaporized nitrogen. This design was vulnerable to a single point of failure. Once the tube became bent, there was nothing else to prevent the release of liquid nitrogen. The design team did not adequately consider the consequences of the failure of the bubbler tube—a critical safety component—and did not identify appropriate safeguards to mitigate its potential failure. In addition, during fabrication of the immersion freezer, Messer failed to detect a manufacturing defect that exacerbated the potential for the bubbler tube to be bent—a missing clamp meant to secure the bubbler tube to the freezer. (Section 4.1)

- **Atmospheric Monitoring and Alarm Systems.** There is abundant industry guidance on the importance of atmospheric monitoring when the potential for hazardous atmospheres exists. FFG, however, did not install atmospheric monitoring equipment in the freezer room. As such, there was no equipment installed to detect the oxygen-deficient atmosphere in the freezer room, automatically shut off the liquid nitrogen supply, and notify personnel to evacuate the area. (Section 4.2)

- **Emergency Preparedness.** FFG did not inform, train, equip, drill, or otherwise prepare its workforce for a release of liquid nitrogen. FFG's workforce lacked knowledge of the hazards of nitrogen and its ability to create an oxygen-deficient atmosphere, were unable to recognize an oxygen-deficient

atmosphere, and lacked any equipment or PPE that would have enabled safe entry into an oxygen-deficient atmosphere. As a result, after the discovery of the first two deceased workers, four other employees who entered the freezer room also died from asphyxiation due to the released nitrogen during response or rescue attempts. Three other employees responded and were seriously injured. At least seven other employees responded and were not seriously injured, but they were at risk of nitrogen asphyxiation due to their proximity to the release. FFG also did not proactively interact with local emergency responders prior to the incident, despite relying upon them to respond to emergencies at its facility. (Section 4.3)

- **Process Safety Management System.** FFG had no documented process safety management policy, allowed the job position responsible for safety management to be vacant for more than a year prior to the incident, did not evaluate the process hazards associated with the freezer, lacked written procedures and a management of change process, and did not train its workers on the asphyxiation hazards of liquid nitrogen. Despite robust industry guidance on process safety management systems, FFG was not required by law to implement it. As a result of the lack of specific regulations for cryogenic asphyxiants, FFG did not implement systems and practices that could have reduced the incident's severity or prevented the accidental release altogether. (Section 4.4)

- **Product Stewardship.** Messer owned the liquid nitrogen bulk storage tanks and the Line 4 immersion-spiral freezer and leased the equipment to FFG. At the time of the incident, Messer had institutional knowledge, experience, policies, and practices for effective product stewardship but applied those practices only to the bulk storage tanks and not to the Line 4 freezer process. Throughout its relationship with FFG, Messer had identified issues with FFG's safety practices and nonconformance to industry guidance, but despite FFG's unsafe practices, Messer continued to supply FFG with liquid nitrogen. Had Messer suspended service until FFG corrected known safety deficiencies, this incident may have been prevented. (Section 4.5)

CAUSE

The CSB determined the cause of the liquid nitrogen release was the failure of the immersion freezer's liquid level control system to accurately measure and control the liquid nitrogen level inside the freezer, which resulted from deformation of the system's bubbler tube component.

Contributing to the incident were 1) Messer's design of the freezer, which allowed the failure of a single level measurement device to result in an uncontrolled loss of containment of liquid nitrogen, 2) FFG's lack of any process safety management systems or practices that could have prevented the incident, 3) a lack of regulatory coverage for liquid nitrogen, which enabled FFG to elect not to implement process safety practices that could have prevented the incident, and 4) Messer's inadequate product stewardship practices, which resulted in Messer continuing to supply FFG with liquid nitrogen despite FFG's unsafe practices.

Contributing to the severity of the incident were 1) FFG's inadequate emergency preparedness, which resulted in at least 14 employees responding to the release by entering the freezer room or the surrounding area to investigate the incident or to attempt to rescue their coworkers, and 2) the absence of atmospheric monitoring and alarm devices that could have alerted workers to the presence of a hazardous atmosphere and warned them against entering it.

The Messer Line 4 liquid nitrogen immersion-spiral freezer[a] process (**Figure 2**) occurred in two stages. In the first stage, a conveyor belt carried the fully cooked diced chicken into an immersion freezer where it was partially frozen in a -320°F (-196°C) liquid nitrogen bath. The chicken then moved through a transition box to the second stage, where a conveyor belt carried the partially frozen diced chicken through the spiral freezer as internal fans circulated vaporized liquid nitrogen within the freezer to complete the freezing process. The frozen chicken exited the spiral freezer and was packaged and shipped to customers.

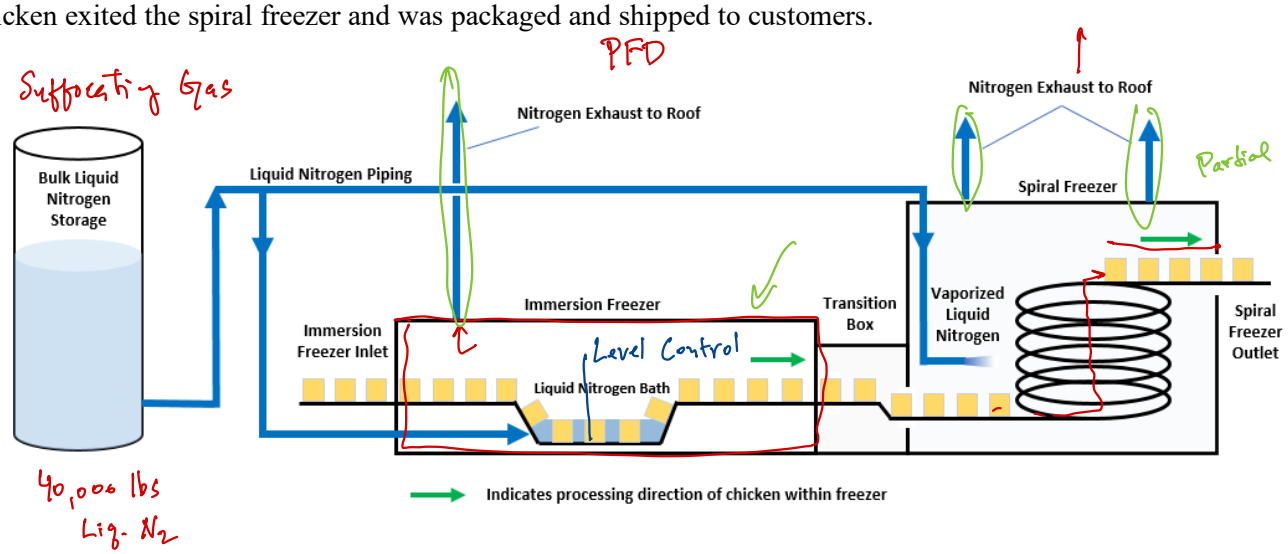


Figure 2. Conceptual diagram of the Line 4 liquid nitrogen freezing system, not drawn to scale. (Credit: Messer, annotations by CSB)

Exhaust systems connected to both the immersion and spiral freezers directed the nitrogen gas from inside the freezers to outside the building through discharge ducting on the roof.

Control valves regulated the flow of liquid nitrogen from storage tanks outside the building through piping along the roof and into each freezer. FFG workers used a computer control system with touch screens, called a Human Machine Interface (HMI), to set operating parameters for the freezer system, such as the liquid nitrogen level inside the immersion freezer, conveyor belt speeds, and exhaust fan speeds. The computer control system programming included safety interlocks to automatically prevent inadvertent, hazardous liquid nitrogen releases, which could be activated either by the control system measuring a high liquid level or by pushing emergency stop (E-stop) buttons located on the freezer exterior. The control system measured the liquid level in the immersion freezer using a device called the bubbler tube, which is discussed in further detail in Section 3.

To enable access to the inside of the immersion freezer, the freezer was also equipped with a lid that could be raised, and the transition box had a door that could be opened. The lid and the door were equipped with permissive proximity switches, or safety switches; as designed, when the lid and door were closed, the safety switches would enable the freezer to operate normally. If the safety switches lost proximity (meaning that the door was open and/or the lid was raised), the freezer system was designed to shut off liquid nitrogen supply to the freezers and to increase the speed of the nitrogen exhaust fans.

[a] The immersion-spiral freezer consisted of two discrete freezers—an immersion freezer and a spiral freezer—connected in series.

1.4 LINE 4 FREEZER ROOM

Figure 3 below illustrates the layout of the FFG Line 4 freezer room.

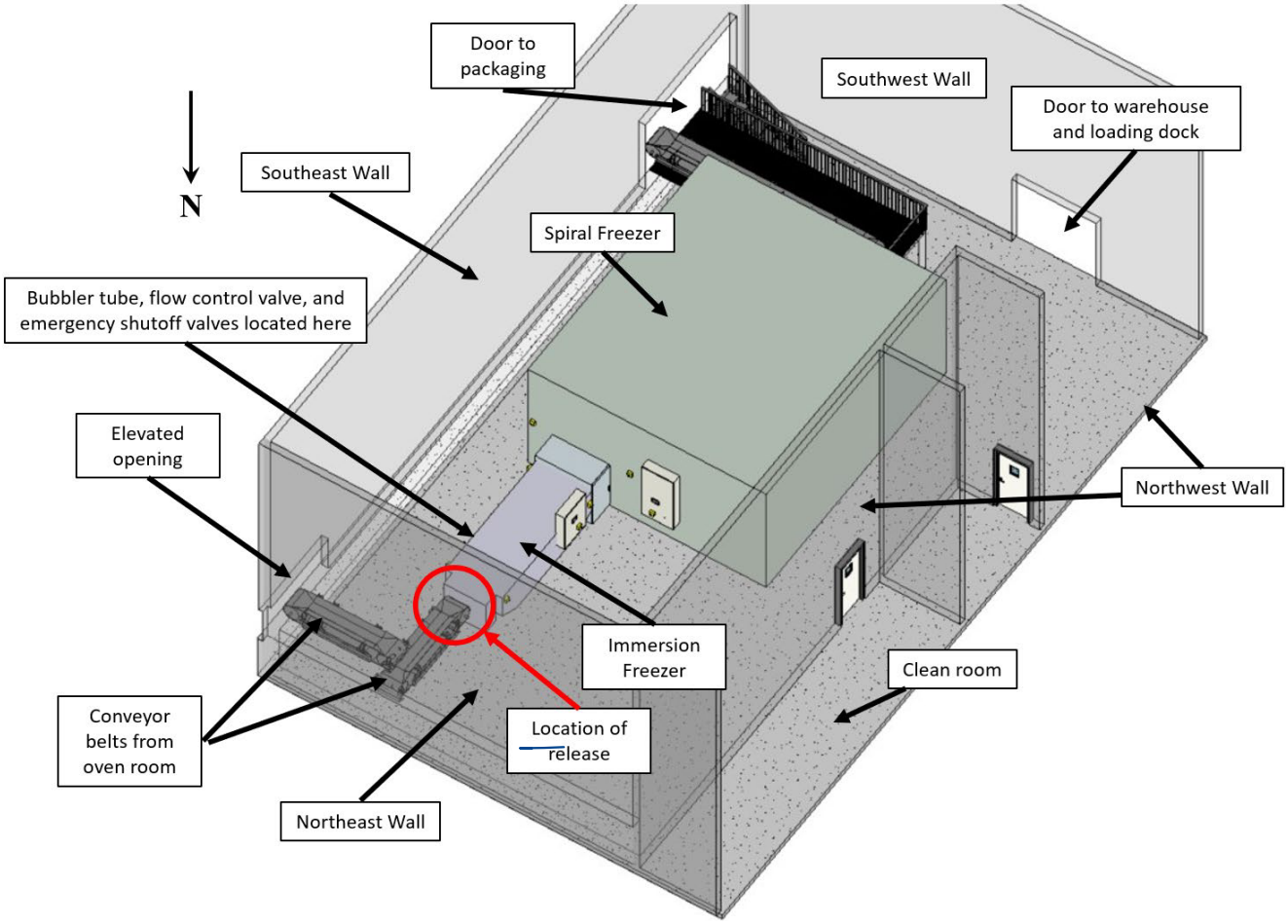


Figure 3. Diagram of the FFG Line 4 freezer room. (Credit: CSB)

In total, the room had four openings. On the southeast wall, there was an elevated opening through which conveyor belts brought product from the oven room into the freezer room. Also on the southeast wall was a large open doorway into the packaging area. Both of these openings on the southeast wall were elevated roughly five feet from the floor of the freezer room. On the southwest wall, there was a large open doorway leading to the warehouse and loading dock area. This doorway contained heavy plastic curtains separating the freezer room from the warehouse and loading dock. On the northwest wall, there was a single door that separated the freezer room from the clean room. There were no openings in the northeast wall.

The rooms adjacent to the Line 4 freezer room on the northeast and southeast sides were elevated roughly five feet above the floor level in the freezer room (**Figure 4**). The adjacent rooms on the southwest and northwest sides were at the same floor elevation as the freezer room. Thus, the room was recessed below adjacent rooms on two sides and level with the adjacent rooms on the two other sides. Ultimately, below the elevation of five feet, the room was very nearly fully enclosed. The only openings in the room below five feet of elevation were a

standard-size door on the northwest wall, and a large doorway on the southwest wall that was partially obstructed by heavy plastic strip curtains.

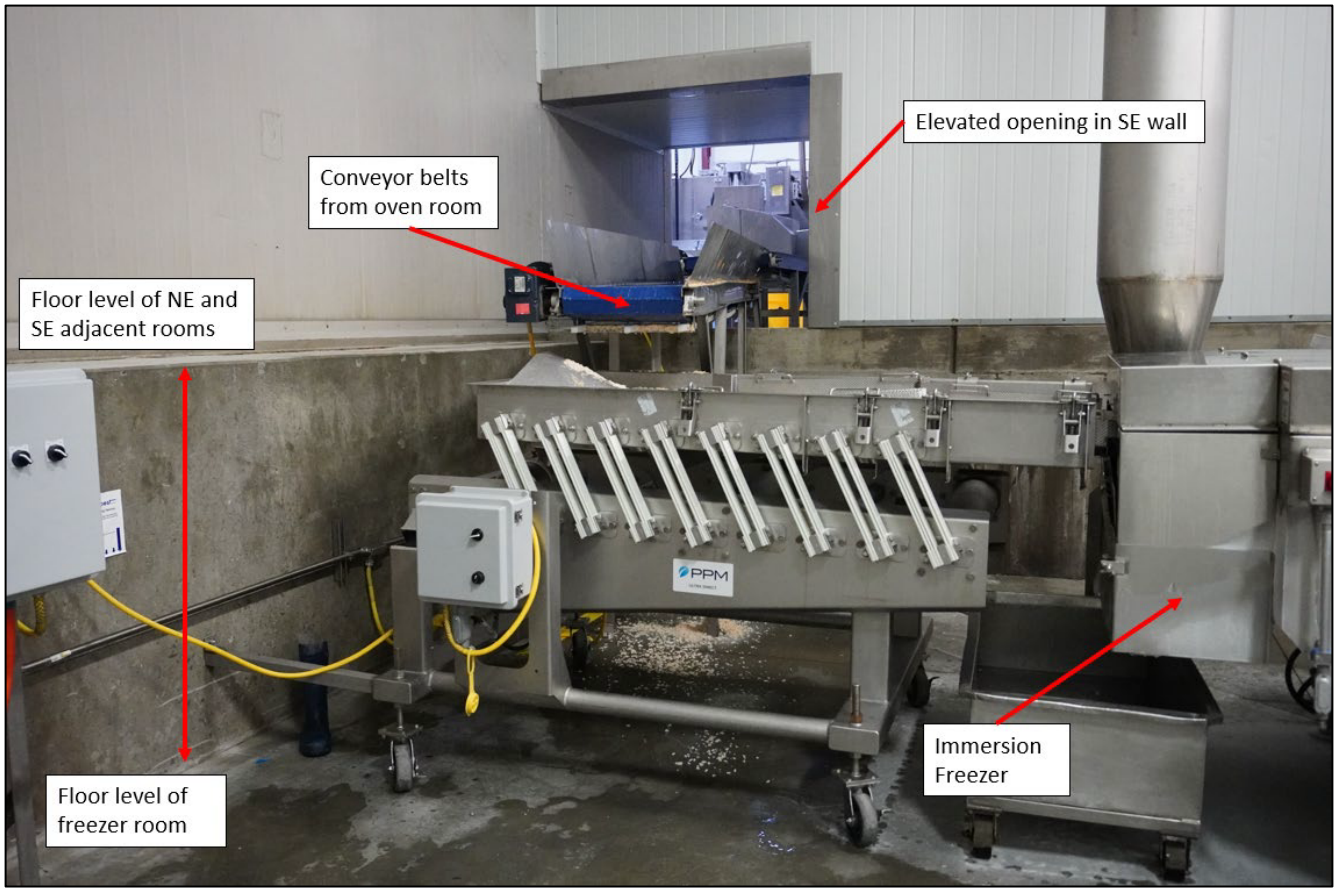


Figure 4. Photo showing difference in elevation between Line 4 freezer room and the adjacent rooms on the northeast and southeast sides. (Credit: CSB)

Air circulated through the room through the four openings described above. The room had no mechanical ventilation or heating, ventilation, and air conditioning (HVAC), although the immersion-spiral freezers contained exhaust systems designed to remove the vaporized nitrogen from the normal operation of the freezers (described previously in Section 1.3 and **Figure 2**).

1.5 Division of Responsibilities Between FFG and Messer

As part of FFG's and Messer's supply agreement, Messer retained ownership[a] of the bulk liquid nitrogen tanks, the Line 2 spiral freezer, and the Line 4 immersion-spiral freezer equipment. Messer also supplied FFG with liquid nitrogen and maintained and inspected the bulk tanks. FFG was responsible for maintaining the equipment site; ensuring the site complied with safety and environmental regulations; providing necessary

[a] FFG leased the equipment from Messer.

denser than air and can settle and collect in low areas [4]. Contact with liquid nitrogen or cold nitrogen gas can cause severe cold burns, frostbite, and hypothermia.[a] Prolonged breathing of extremely cold gas may damage lung tissue.

1.6.2 ASPHYXIATION

Liquid nitrogen has a liquid-to-gas expansion ratio of 1 to 696 at 70°F (21°C), meaning that nitrogen gas will expand to fill the volume of a space 696 times greater than its volume when in the liquid phase [5, p. 5]. High concentrations of nitrogen gas in an enclosed area will displace oxygen, creating an oxygen-deficient atmosphere. Atmospheres containing less than 19.5% oxygen are considered oxygen-deficient and can cause a range of effects on the human body, as shown in **Figure 6**. Atmospheres containing very low oxygen concentrations can lead to asphyxiation, an effect in which the body receives inadequate oxygen, resulting in loss of consciousness or death [6].[b] At these very low oxygen concentrations (i.e., less than 10%), exposure may result in the rapid loss of consciousness without warning [7], which can occur in as few as one or two breaths [6, p. 4].

Atmospheric Oxygen Concentration (%)	Possible Results
20.9	Normal
19.0	Some unnoticeable adverse physiological effects
16.0	Increased pulse and breathing rate, impaired thinking and attention, reduced coordination
14.0	Abnormal fatigue upon exertion, emotional upset, faulty coordination, poor judgment
12.5	Very poor judgment and coordination, impaired respiration that may cause permanent heart damage, nausea, and vomiting
<10	Inability to move, loss of consciousness, convulsions, death

Figure 6. Effects of oxygen deficiency on the human body [8, p. 3]. (Credit: CSB)

1.7 LIQUID NITROGEN IN FOOD PROCESSING

Cryogenic freezers using inert gases, such as liquid nitrogen, have become common in the food processing industry due to their relatively low capital investment, smaller size, and faster freezing time compared with

[a] Symptoms of hypothermia include a slowing down of physical and mental responses, unreasonable behavior, speech or vision difficulty, stumbling, and cramps and shivers.

[b] Typically, air contains about 21% oxygen [52]. Symptoms of asphyxia may include rapid breathing, nausea, vomiting, inability to move, convulsive movements, collapse, abnormal pulse, rapid fatigue, faulty judgment, insensitivity to pain, and abnormal emotions.

ammonia-based mechanical freezers [9]. In addition to Messer, several companies manufacture cryogenic freezers for the food processing industry comparable to the one discussed in this report.

Cryogenic immersion-spiral freezers are used in various food freezing applications, including the freezing of poultry, beef, diced luncheon meats, pizza toppings, marinated meats and vegetables, shrimp, and seafood fillets [10].

Within the United States, at least 220 poultry processing facilities handle cryogenic liquids. In June 2021, the U.S. Poultry & Egg Association polled poultry processing facilities and found that 74% of the respondents use cryogenic freezers, with 22% using liquid nitrogen, 33% using carbon dioxide, and 19% using both liquid nitrogen and carbon dioxide, as shown in **Figure 7**.

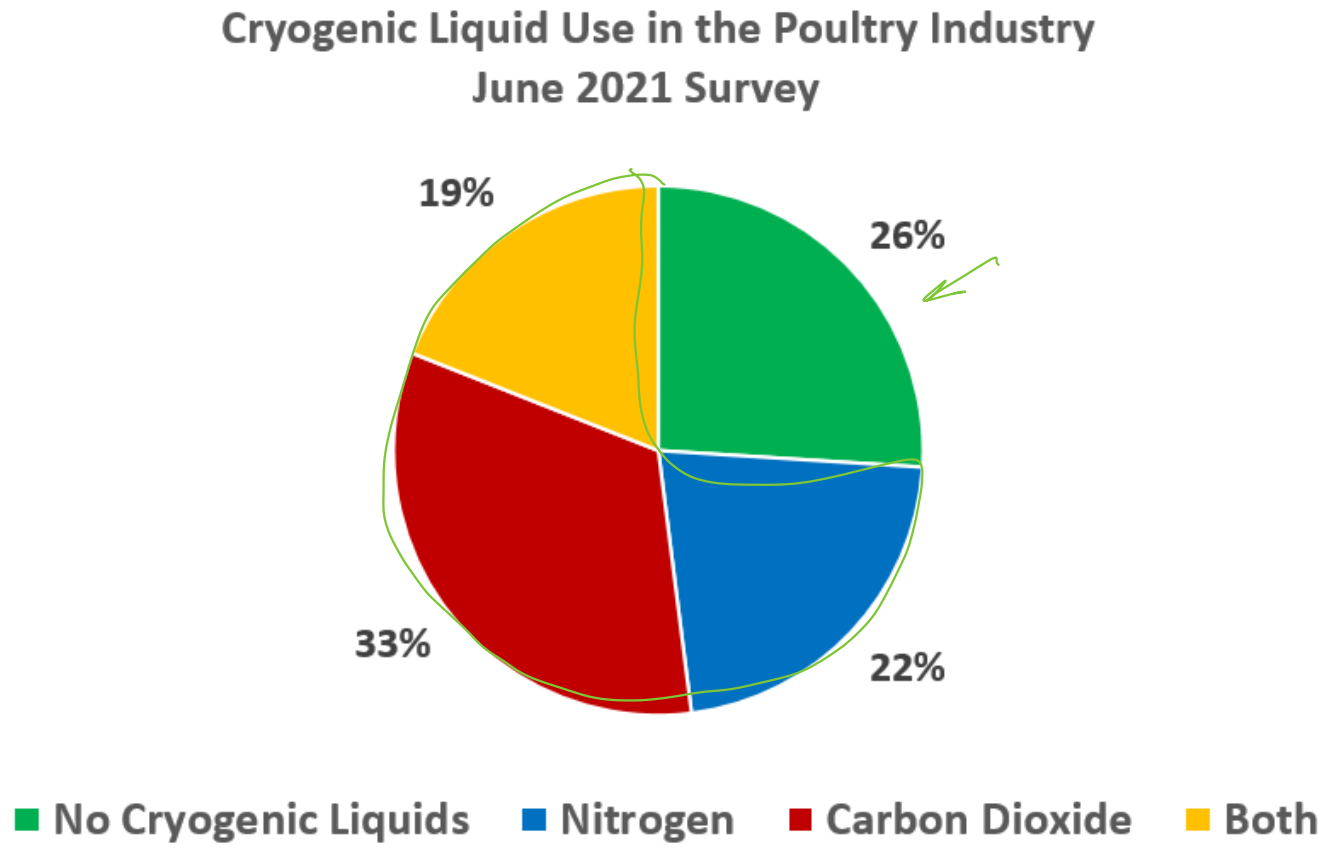


Figure 7. Cryogenic use in the poultry industry. (Credit: U.S. Poultry & Egg Association, formatted by CSB)

1.8 Regulatory Coverage

The U.S. Occupational Safety and Health Administration (OSHA) and the U.S. Environmental Protection Agency (EPA) do not define liquid nitrogen as a highly hazardous chemical or extremely hazardous substance. Consequently, OSHA's Process Safety Management (PSM) standard (29 CFR 1910.119) and the EPA's Risk Management Program (RMP) rule (40 CFR 68) did not apply to FFG's liquid nitrogen process [11, 12]. Although FFG also utilized anhydrous ammonia (which is regulated by PSM and RMP) in processes separate from the Line 4 liquid nitrogen process, FFG asserted that its ammonia processes were also not covered under

Also, beginning two days prior to the incident, FFG experienced issues when the immersion-spiral freezer was not fully freezing the diced chicken product. FFG employees stopped Line 4 production operations multiple times to troubleshoot the problem. On January 26, 2021, Messer was on-site to witness FFG's attempt to process the diced product. Based on observations that day, the liquid nitrogen level in the immersion freezer was lowered to avoid uneven belt loading, which Messer suspected was causing the freezing issues.

The next day, on January 27, 2021, FFG continued to experience freezing issues. FFG increased the liquid level in the immersion freezer in an attempt to resolve the problem, but this action did not resolve the issue. Subsequently, an FFG maintenance worker reported to management that he believed the liquid nitrogen level was too low. As a result, the manager called Messer for guidance. Messer recommended that FFG further increase the nitrogen level in the freezer and provided FFG with the username and password required to modify this parameter in the system. Production on the immersion-spiral freezer resumed that afternoon and continued through the end of the workday.

2.2 Day of Incident

2.2.1 Liquid Nitrogen Release

On January 28, 2021, operations on Line 4 commenced at approximately 7:16 a.m. Continuing the trend from the previous days, soft, partially-frozen chicken product was observed coming out of the immersion-spiral freezer. As a result, Line 4 stopped processing chicken at approximately 8:14 a.m. so that maintenance personnel could troubleshoot the issue. At approximately 8:20 a.m., the Line 4 Packaging Supervisor[a] instructed the Line 4 workers to go on break while maintenance workers attempted to resolve the freezing issue. The maintenance workers conducted the maintenance without shutting down the freezer and with the transition box door and immersion freezer lid safety switches intentionally bypassed.

The U.S. Chemical Safety and Hazard Investigation Board (CSB) determined that, likely while the Line 4 workers were on break, the bubbler tube in the immersion freezer somehow became bent, thereby preventing the freezer's liquid nitrogen level control system from working properly (described further in Section 3). The liquid nitrogen level then increased in an uncontrolled manner, and liquid nitrogen overflowed from the immersion freezer, filling the freezer room with vaporized nitrogen that displaced the room's oxygen.

After about 20 minutes, at approximately 8:40 a.m., the Line 4 workers returned from break. At this time the workers noticed a cloud of white fog coming from the Line 4 freezer room but took no further action. According to one employee, because workers regularly observed white fog in the freezer room, they did not believe there was a problem. By 9:40 a.m., having not heard from the Line 4 Packaging Supervisor for over an hour, one of the Line 4 workers decided to look for him. The Line 4 worker entered the Line 4 freezer room through an elevated opening,[b] pictured in **Figure 3** and **Figure 9**.

[a] The Line 4 Packaging Supervisor was later found fatally injured.

[b] Chicken enters the freezer room via conveyor belt through this opening. The bottom edge of this opening is elevated approximately 4 feet and 9 inches above the freezer room floor. This opening was not intended for human use as a passageway and was intended only to allow for the conveyor belts to enter the freezer room, and to allow for some make-up ventilation. The Line 4 worker crouched down and crawled through the opening.

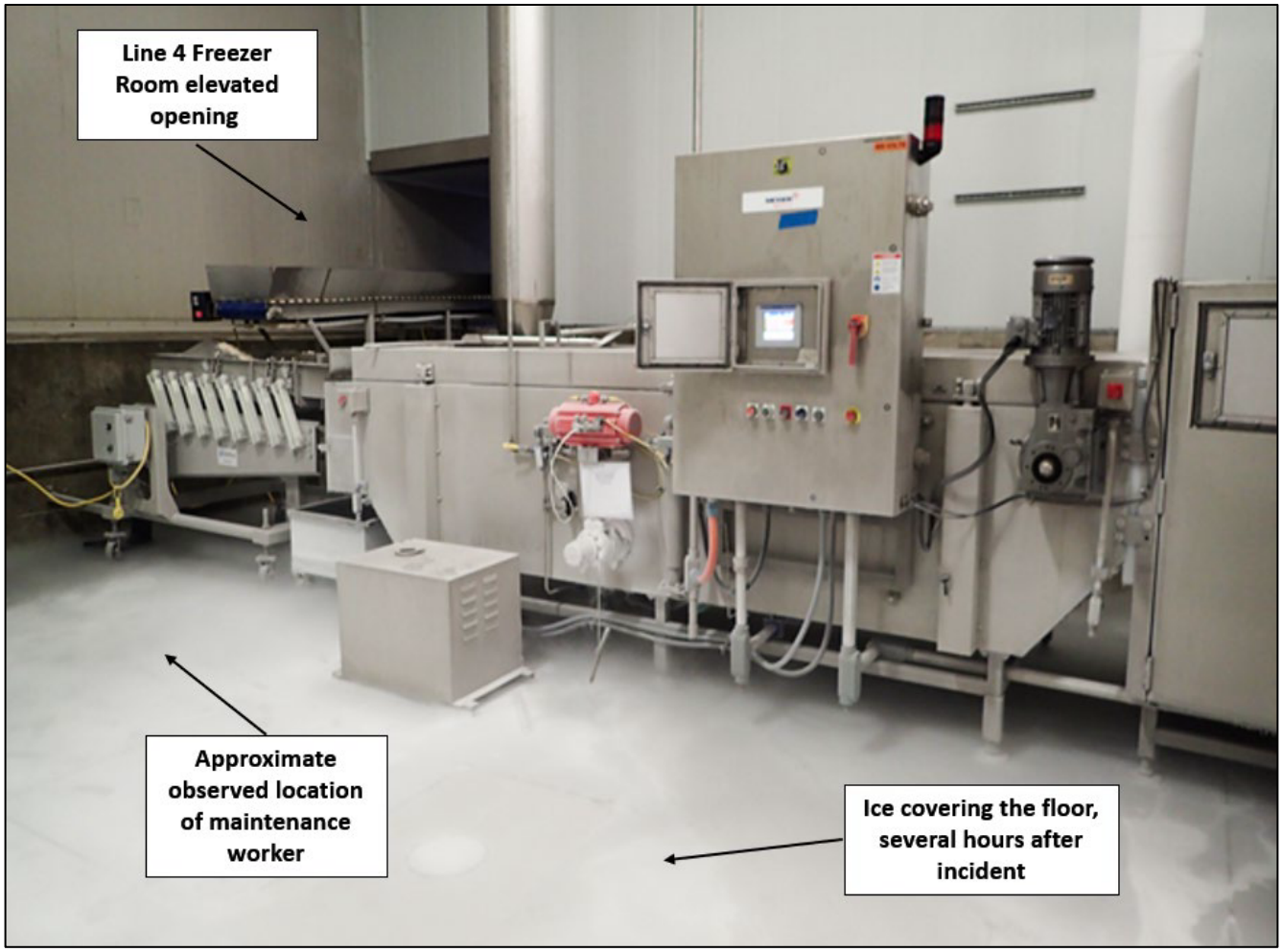


Figure 9. Line 4 immersion-spiral freezer in Line 4 freezer room several hours after the incident. (Credit: Hall County Fire Services, annotations by CSB)

The Line 4 worker observed a dense white cloud approximately four feet high filling the room.[a] She also saw one of the maintenance workers lying motionless on the ground in front of the conveyor belt. The Line 4 worker exited the freezer room by crawling back through the elevated opening and returned to where the other Line 4 workers were congregated in the RTE area. She informed the other workers of what she saw and then crawled through the elevated opening again, this time with another worker to confirm her observations. The Line 4 workers again returned to the area where the other workers were congregated and then left to locate the Line 1 Packaging Supervisor to inform him about what they had observed.

2.2.2 Incident Response

Because FFG did not have a surveillance system, the only evidence available to the CSB to use to reconstruct response activities were the firsthand accounts of the employees who were not fatally injured during the

[a] Cold nitrogen gas could create a fog cloud of condensed moisture in the air. This white cloud may obscure visibility [7].

3 TECHNICAL ANALYSIS

After the incident, the CSB examined the immersion freezer. Part of the examination involved raising the freezer lid and conducting a visual examination inside the freezer. The CSB found the bubbler tube bent, as shown in **Figure 10**.



Figure 10. Immersion freezer bubbler tube, as found during the post-incident examination of the freezer. As designed, the tube should be pointing straight downward. (Credit: CSB)

As discussed previously, the immersion freezer was designed to submerge product in a bath of liquid nitrogen, maintaining a user-specified liquid height in the bath. Messer's predecessor, Linde, designed the immersion freezer with a liquid level measurement device called a bubbler tube.[a] As shown in **Figure 11**, a bubbler tube works by maintaining a constant flow rate of vapor through the tube and into the liquid and measuring the differential pressure between the liquid and atmospheric pressure [11, p. 113]. The immersion freezer control system then used that differential pressure measurement to calculate the liquid level in the bath and would change the position of the liquid nitrogen flow control valve to maintain the liquid level at the user-specified setpoint. Messer also equipped the freezer with a high-level safety interlock that when activated was designed to

[a] Also referred to as a "dip tube" [11, p. 113].

close valves to shut off the flow of additional liquid nitrogen into the freezer, shown below in **Figure 12**. The level control loop and the high-level safety interlock both used the bubbler tube as the input sensor.

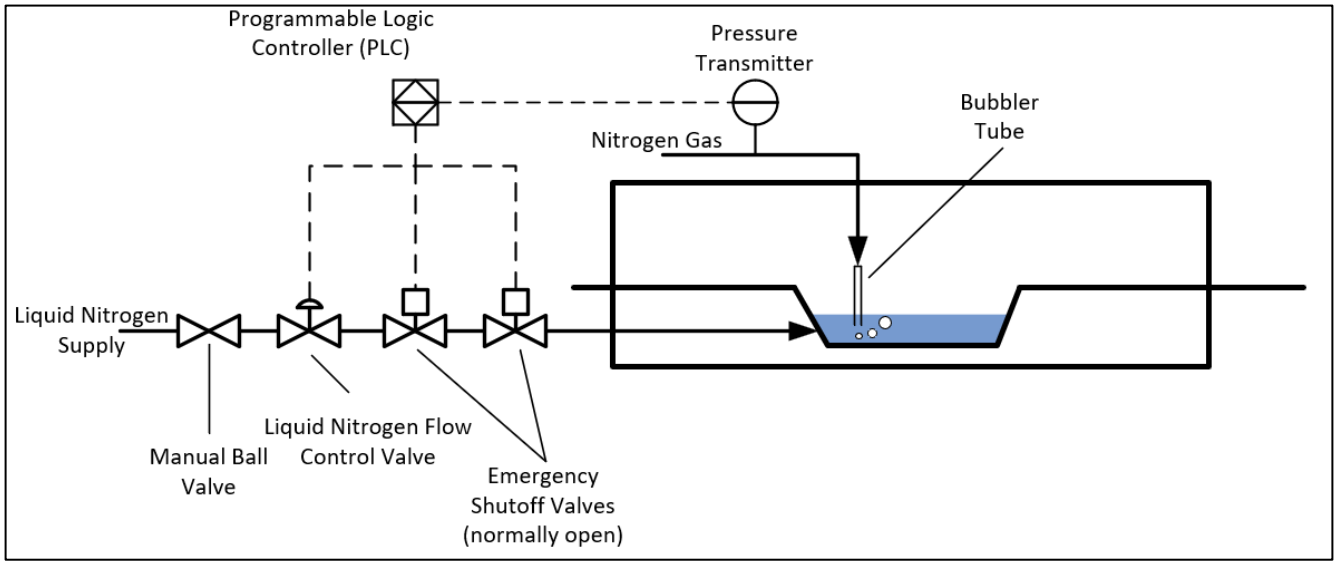


Figure 11. Simplified conceptual diagram of the immersion freezer level control and high-level safety interlock system. Not drawn to scale. (Credit: CSB)

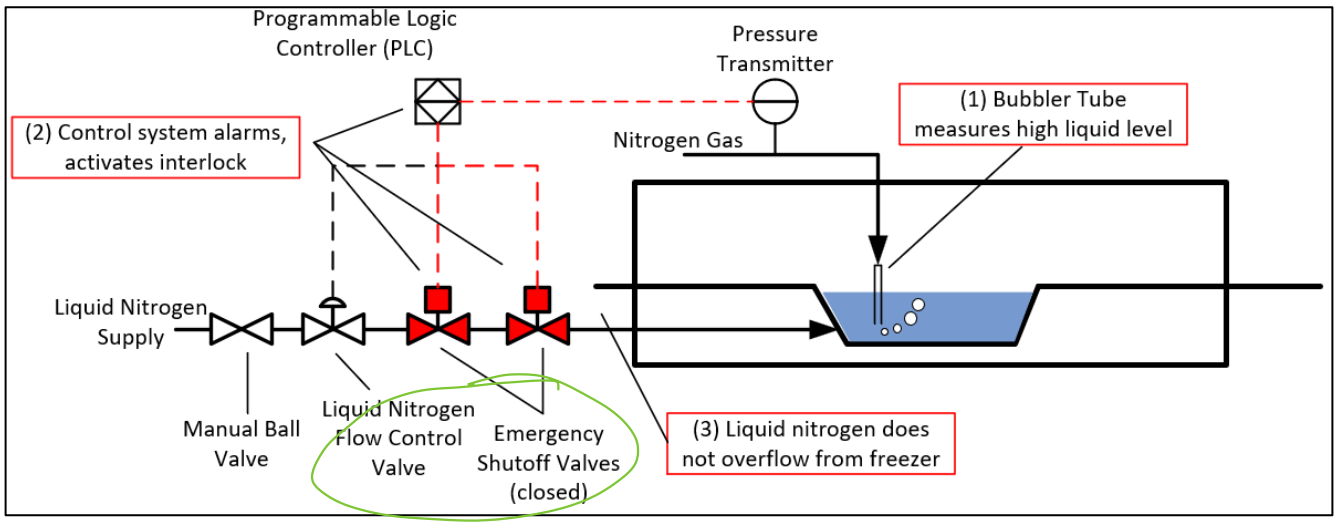


Figure 12. Simplified conceptual diagram showing the normal function of the immersion freezer high-level safety interlock. Not drawn to scale. (Credit: CSB)

The CSB tested the functionality of the immersion freezer level control system after the incident. The test involved filling the immersion tub with water, with the intent of determining 1) whether liquid would overflow from the freezer, 2) the location from which the liquid would overflow, and 3) the position of the tip of the bent bubbler tube relative to the overflow liquid level in the tub.

During the test, the CSB documented the following conditions, pictured below in **Figure 13** and illustrated in **Figure 14**:

- Water overflowed from the inlet side of the freezer.
- The overflow liquid level was lower than the elevation of the tip of the bent bubbler tube.
- The freezer level control system measured no liquid level during the test, and the system called for the maximum flow of liquid for the duration of the test.

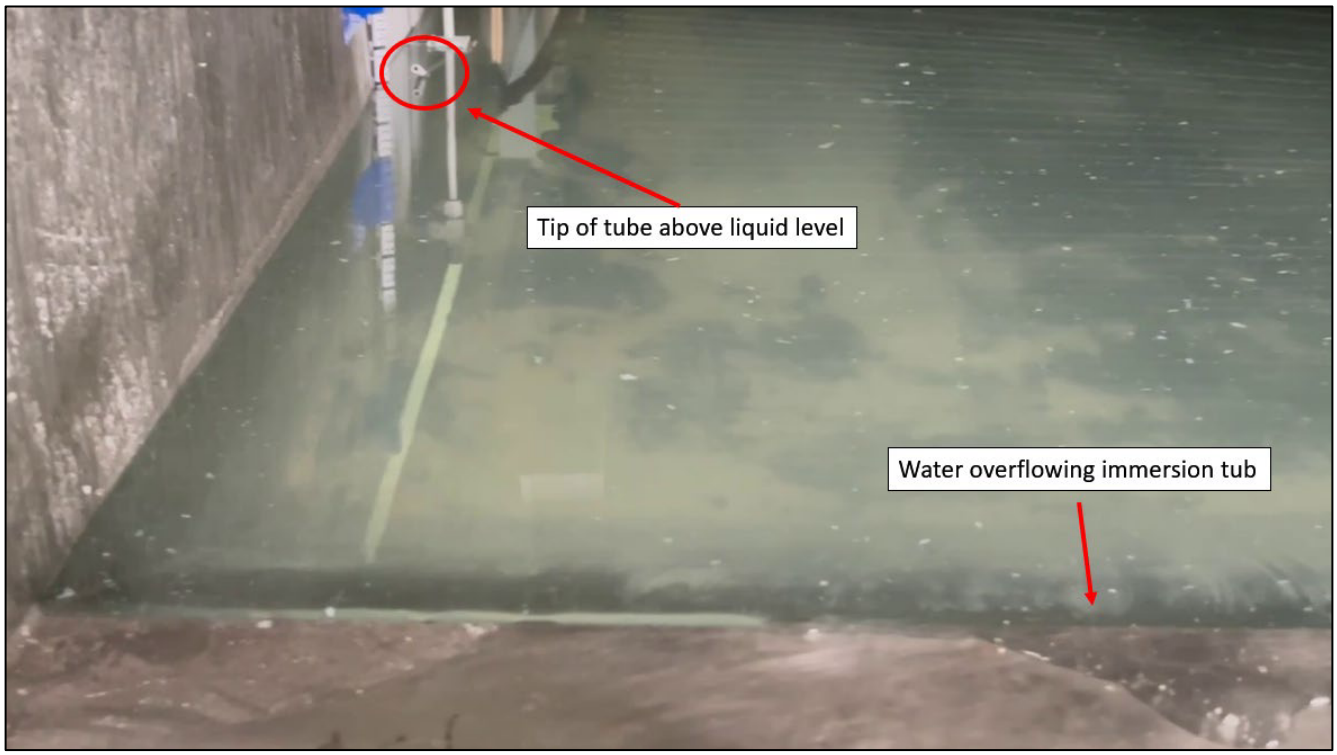


Figure 13. Water overflowing from the immersion freezer tub during post-incident overflow test. (Credit: CSB)

During the incident, because the tip of the tube was bent above the overflow level, the level control system would have incorrectly measured no liquid level and would have continued calling for additional liquid nitrogen even though the immersion tub overflowed. Also, because the level measurement would have been artificially low, the high-level interlock would not have functioned correctly (as illustrated in **Figure 14**). The CSB examined the freezer alarm and event log and confirmed that no high-level event was recorded by the freezer control system during the incident. The CSB concludes that liquid nitrogen overflowed from the immersion freezer. The liquid nitrogen overflow was caused by the failure of the freezer's level control and high-level safety interlock systems.

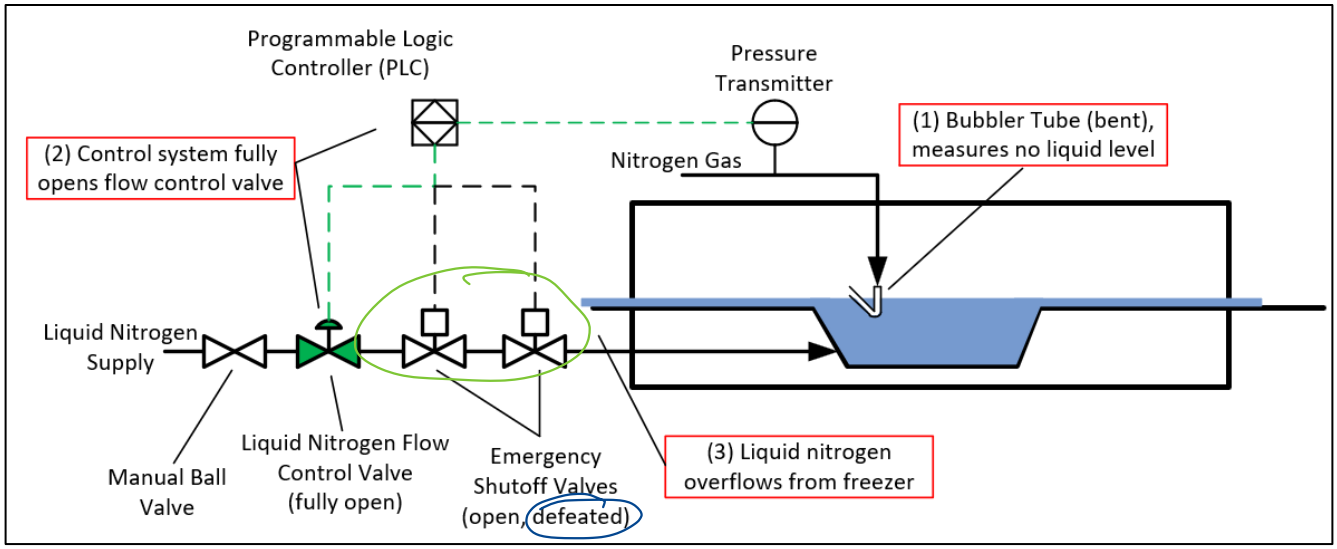


Figure 14. Simplified conceptual diagram showing the failure state of the immersion freezer level control system. Not drawn to scale. (Credit: CSB)

The CSB conducted a second overflow test using an as-designed exemplar bubbler tube in place of the bent incident tube. Both the level control and high-level interlock systems operated as designed (as shown in **Figure 12**), indicating that during the incident there was likely nothing preventing the correct function of any of the level control or interlock equipment other than the bent bubbler tube. The CSB concludes that the failure of the freezer's level control and high-level safety interlock systems was caused by the deformation of the bubbler tube.

The immersion freezer level control system operated as designed (and the freezer did not overflow) in the days leading up to the incident including up until 8:14 a.m. on the morning of the incident when production was stopped and the freezer was turned over to maintenance. Therefore, the CSB concludes that the bubbler tube was likely bent on the morning of the incident during maintenance troubleshooting activities, likely between 8:20 a.m. and approximately 9:30 a.m. However, there was insufficient evidence to determine exactly when the tube was bent. Therefore, the CSB was not able to determine precisely when the uncontrolled release began.

There was insufficient evidence to determine how the bubbler tube was bent. The CSB considered several competing hypotheses, including 1) intentional or unintentional human manipulation of the tube by hand or using various tools, 2) frozen product impacting the tube and bending it over time, and 3) a mass of frozen product becoming wedged between the tube and the wall of the immersion freezer or the conveyor belt.

Various hypotheses were tested extensively by the CSB, Messer, and FFG. Though several of the tests produced results similar to the as-found condition of the incident tube, none of the tests reproduced the exact state in which the incident tube was found (such as the degree of the bend, the direction of the bend, or the presence or absence of certain "witness marks," which are specific impressions such as scratches and dents left in the metal). As a result, the CSB was not able to definitively determine how the tube was bent.

There was also insufficient evidence to determine the actions that the two maintenance workers took during their troubleshooting efforts, or the sequence of events that took place in the freezer room between when the freezer

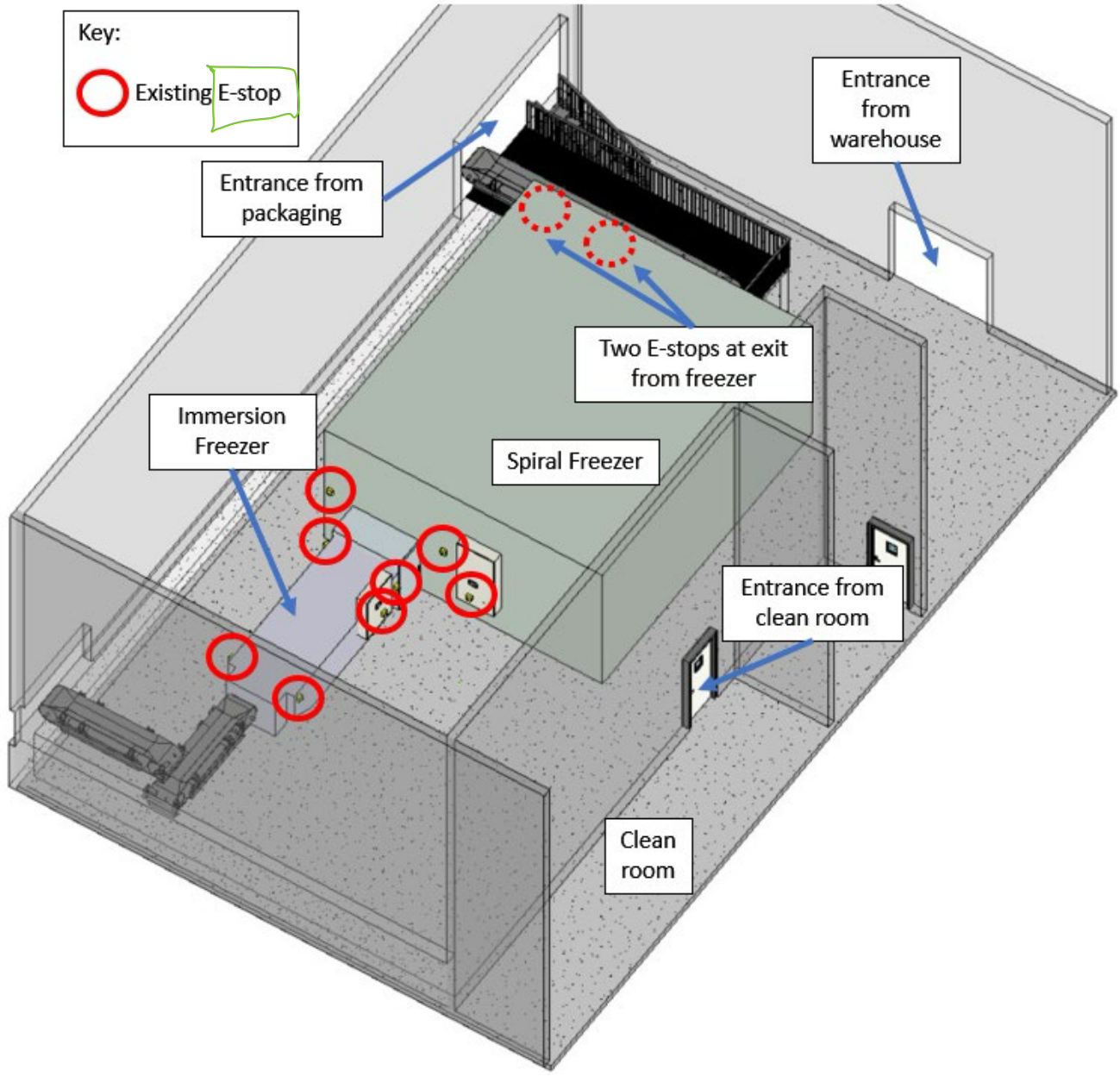


Figure 20. Locations of the Line 4 immersion-spiral freezer E-stops at the time of the incident. (Credit: CSB)

As shown, the 10 E-stop buttons consisted of five buttons on each freezer; one on either side of the inlet and outlet of each freezer, and one on each HMI panel. All 10 buttons were located within the freezer room. The result of this placement was that activation of any of the E-stop buttons required someone to approach the source of the release to activate an E-stop. As stated earlier, FFG had no employees equipped for or trained in emergency response, and as a result, after the release began, no one at the company could safely activate any of the E-stops.[a] The only way that anyone at the company could safely shut off the flow of liquid nitrogen during

[a] As discussed above, the two FFG maintenance workers may have had the opportunity to activate an E-stop and avert the release, but it was impossible to determine whether the two workers were aware of the bent bubbler tube, whether they had sufficient knowledge and training to understand what hazard the bent tube presented, or to what extent the workers were aware of the impending overflow.

4.4.1.8 Conclusions

The CSB concludes that FFG lacked an effective process safety management system to identify, evaluate, and control the hazards of the liquid nitrogen process. The lack of safety oversight by FFG leadership, absence of a systematic process hazard analysis, lack of written procedures, lack of any management of change practices, and failure to communicate hazards resulted in the unmitigated handling of a cryogenic asphyxiant by untrained and unprepared personnel. This incident could have been prevented had FFG practiced robust process safety management.

As discussed in Section 1.1.1, Gold Creek currently owns and operates the former FFG facility. As of this report's publication, there are currently no liquid nitrogen freezing processes at the former FFG Plant 4 building that Gold Creek now operates. Therefore, the CSB does not issue a recommendation to Gold Creek related to process safety management of liquid nitrogen processes.

KEY LESSON

Food manufacturers are not immune from chemical hazards and process safety risks. Whenever an organization introduces a hazardous chemical into its process, it should implement robust process safety management practices to effectively control the risks, regardless of whether any regulation requires the organization to do so.

4.4.2 Lack of Regulations for Cryogenic Asphyxiants

As discussed in Section 4.4, the FFG liquid nitrogen freezer process was not subject to OSHA's PSM standard (or EPA's RMP rule). In addition, there is no existing regulation that requires organizations that handle or process cryogenic asphyxiants to adhere to the requirements of CGA, CCPS, or any safe handling and safety management system guidelines.

4.4.2.1 Fatalities from Liquid Nitrogen Release Incidents

By OSHA's definition, a highly hazardous chemical is "a substance possessing toxic, reactive, flammable, or explosive properties" [26].[a] Since liquid nitrogen is not toxic, reactive, flammable, or explosive, it does not meet OSHA's definition of a highly hazardous chemical.[b] Regardless, uncontrolled liquid nitrogen releases have resulted in serious injuries and fatalities in several industries. Since 2008, the CSB is aware of at least seven incidents involving liquid nitrogen that have resulted in 13 fatalities in the United States, as shown in **Table 4**. None of these incidents were subject to a regulated process safety management standard for liquid nitrogen.

[a] 29 CFR 1910.119(b)

[b] Liquid nitrogen meets the CSB's definition of an "extremely hazardous substance," which is defined in 40 CFR 1604.2 as "any substance which may cause death, serious injury, or substantial property damage […]."

Table 4. Liquid nitrogen asphyxiation fatalities since 2008

Company	Location	Date of Incident	Severity	Industry
Foundation Food Group	Gainesville, Georgia	January 28, 2021	6 fatalities	Food Manufacturing
California Ranch Food Company	Vernon, California	December 1, 2020	2 fatalities	Food Manufacturing
Custom Genetic Solutions, LLC	Mitchell, South Dakota	November 20, 2019	1 fatality	Genetics
XYTEC	Augusta, Georgia	February 5, 2017	1 fatality	Genetics
ATI Allvac	Richburg, South Carolina	June 26, 2012	1 fatality	Metalworking
Blowout Tools, Inc.	Fannin, Texas	October 30, 2010	1 fatality	Equipment Manufacturing
Blommer Chocolate Company	Chicago, Illinois	June 8, 2008	1 fatality	Food Manufacturing

The liquid nitrogen asphyxiation incidents occurring at California Ranch Food Company (California Ranch) and FFG collectively resulted in eight fatalities within less than two months.

4.4.2.2 Existing Regulations and Guidance

OSHA's stated mission is to "ensure safe and healthful working conditions for workers by setting and enforcing standards and by providing training, outreach, education and assistance" [29]. Since OSHA's standards cover FFG and its workers, the violations cited by OSHA at the FFG facility prior to and following the January 2021 incident provide insight into the scope of applicable existing regulations.

OSHA Citations Prior to the FFG Incident

Between 2016 and 2020, OSHA inspected the FFG facility three times, as shown in **Table 5**. While OSHA found several violations, OSHA did not document any findings or concerns relating to FFG's handling of liquid nitrogen during these visits.

Table 5. OSHA inspection citations at FFG between 2016 and 2020

Inspection Date	Citation(s)	Violation (Paraphrased)
December 10, 2020	General Duty Clause Section 5(a)(1)	Workers were struck by 40-pound blocks of frozen chicken traveling on a conveyor.
	Control of Hazardous Energy 29 CFR 1910.147	FFG did not inspect equipment-specific procedures, nor train employees, exposing workers to amputation hazards.

5 CONCLUSIONS

5.1 FINDINGS

Technical Analysis

1. Liquid nitrogen overflowed from the immersion freezer. The liquid nitrogen overflow was caused by the failure of the freezer's level control and high-level safety interlock systems.

2. The failure of the freezer's level control and high-level safety interlock systems was caused by the deformation of the bubbler tube.

3. The bubbler tube was likely bent on the morning of the incident during maintenance troubleshooting activities, likely between 8:20 a.m. and approximately 9:30 a.m.

4. There was insufficient evidence to determine exactly when the tube was bent. Therefore, the CSB was not able to determine precisely when the uncontrolled release began.

5. The CSB was not able to definitively determine how the tube was bent.

6. The CSB could not determine whether the two maintenance workers had sufficient awareness, time, and ability to avert the release by activating an E-stop or to escape the room safely prior to the loss of containment.

7. The uncontrolled release of liquid nitrogen likely ceased at approximately 10:15 a.m. when the bulk storage tank manually-operated discharge valves, located outside the building, were closed.

8. At most, approximately 6,300 gallons (approximately 42,400 pounds) of liquid nitrogen released from the FFG Line 4 immersion freezer, though the actual released quantity was likely less. There was insufficient evidence to determine the exact quantity of released liquid nitrogen because it was not possible to determine when the release began.

9. The released liquid nitrogen vaporized, collected in the freezer room, and produced an oxygen-deficient atmosphere inside the room.

Single Point of Failure

10. The immersion freezer was designed such that the failure of a single level measurement device could defeat both the nitrogen level control system and the emergency interlock intended to stop nitrogen flow to the freezer. After the bubbler tube was bent, there was nothing to prevent the nitrogen release from the freezer.

11. Linde did not design the immersion-spiral freezer in accordance with industry guidance regarding single points of failure for instrumented systems. Had Linde or Messer included additional independent safeguards to protect against overflow events, this incident could have been prevented.

12. Linde did not adequately consider the failure of the liquid nitrogen level control system and did not identify appropriate safeguards to mitigate the potential failure. In essence, Linde incorrectly identified the bubbler tube as a safeguard for itself.

13. Linde did not identify specific incident scenarios or cause-consequence pairs when conducting the PHA for the immersion-spiral freezer, which resulted in Linde's failure to identify adequate safeguards to protect against the overflow of liquid nitrogen from the immersion freezer.

14. Messer's Quality Control procedures and practices were ineffective in ensuring that the two support clamps on the bubbler tube were in place at the time of inspection. As a result, Messer failed to identify the missing support clamp during Quality Control inspection.

15. The combination of unsecured tubing length and support clamp location worsened the potential for the bubbler tube to become bent, which ultimately resulted in the bubbler system becoming non-functional.

Atmospheric Monitoring and Alarm Systems

16. Messer informed FFG of the need for atmospheric monitoring of its liquid nitrogen processes on at least three occasions. Despite Messer's recommendations, neither FFG nor Messer took action to install monitoring or alarm equipment on the Line 4 process, which could have alerted workers to the presence of an oxygen-deficient atmosphere, and, if designed accordingly, could have triggered an emergency shutdown of the liquid nitrogen systems.

17. FFG did not follow industry guidance concerning the use of atmospheric monitoring and alarms for its liquid nitrogen process, and as a result, many personnel were unaware that the freezer room was unsafe to enter on the day of the incident.

18. Had Messer or FFG properly considered, designed, installed, tested, and maintained an atmospheric monitoring and alarm system in the freezer room, workers would have been warned against entering the oxygen-deficient atmosphere, which could have prevented the subsequent fatalities and serious injuries to FFG workers.

19. Industry guidance for carbon dioxide is extensive. Comparable, specific guidance for liquid nitrogen could help prevent future incidents similar to the one at FFG.

Emergency Preparedness

20. Although FFG had a written emergency action plan, it was severely inadequate to address a liquid nitrogen emergency. Its inadequacies included 1) that it was not written in Spanish, the primary language of many of FFG's workers; 2) that it made no mention of the existence of liquid nitrogen at the facility; 3) that it made no mention of the hazards of liquid nitrogen; 4) that it had no instructions for how, whether, or when to respond to a release of liquid nitrogen other than general evacuation instructions; 5) that it contained no information or plan for how employees were to be notified of an emergency; 6) that it contained no information on what constituted an emergency or what types of emergencies to which employees might need to respond; and 7) that it had no provision for proactively interacting with local emergency responders despite the company's stated practice of relying on them for emergency response.

21. As a result of FFG's poor emergency communication, employees attempting to respond to the incident or evacuate the building were only minimally informed, if at all, of the nature and severity of the emergency.

22. The four workers who subsequently were fatally injured were attempting some sort of response to the release.

23. FFG did not prepare its workforce in any meaningful way to respond to a release of liquid nitrogen. The company's deficiencies included 1) its lack of emergency response training for its workforce, 2) its lack of employee training on how to identify a liquid nitrogen release, 3) the lack of automated means to detect and inform its workers of a liquid nitrogen release, and 4) insufficient direction to its employees not to respond or attempt rescue during a liquid nitrogen release.

24. As a result of FFG's unpreparedness, the severity of the incident was greatly increased during evacuation and response activities when four additional employees were fatally injured, three employees were seriously injured, and at least seven others sustained minor injuries or were uninjured but easily could have been seriously or fatally injured. Had FFG effectively prepared its workforce for a liquid nitrogen release, the four additional fatalities and three of the four serious injuries could have been prevented.

25. Gold Creek's policy documents and emergency procedures are more robust than FFG's emergency action plan was at the time of the incident and could have reduced the severity of this incident had FFG implemented similar policies and procedures.

26. The placement of the Line 4 immersion-spiral freezer E-stop buttons required a responding employee or person not otherwise in the freezer room to enter an oxygen-deficient atmosphere during a release of liquid nitrogen to activate an E-stop. This design was unsafe. As a result, once FFG's two maintenance workers became incapacitated, the uncontrolled liquid nitrogen release could not be safely stopped until employees manually closed valves at the bulk liquid nitrogen storage tanks outside of the building, or until emergency responders equipped for entry into an oxygen-deficient atmosphere could enter the freezer room and activate an E-stop. Safer placement of E-stop buttons, and effective employee training on their use, might have helped prevent the death of some or all of the four employees who perished during emergency response.

27. Had FFG and Messer installed E-stop buttons outside the Line 4 freezer room, such as at the entrances to the room or at the bulk storage tanks, the uncontrolled release could have been stopped more expediently.

Process Safety Management System

28. For several months prior to the incident, FFG's safety management organization did not include employees with direct responsibility over the safety practices. Had FFG established process safety programs and objectives; assigned qualified personnel clear responsibilities over process safety management; defined, implemented, and tracked process safety objectives; and ensured organization leaders demonstrated commitment to process safety principles, FFG could have prevented the incident.

29. FFG did not identify or evaluate hazards specific to its process, such as the liquid nitrogen overflow and asphyxiation risks, and did not implement effective controls to mitigate the risk.

30. FFG did not have written procedures to operate or maintain the liquid nitrogen freezer, and therefore the FFG employees were not provided with clear instructions and precautions in operating the equipment. Had FFG developed clear written procedures, it is likely that the FFG workers would have understood the function of the freezer, the importance of critical components, and proper precautions when operating and troubleshooting the equipment, which may have prevented the release.

31. FFG did not have a management of change process to identify, assess, and manage risk introduced by process changes. During the introduction of the liquid nitrogen immersion-spiral freezer, FFG did not consider how the change could impact process safety, did not include approval authorities with explicit responsibilities for safety, and did not address conformance with industry guidance. As a result, FFG did not manage the risks associated with using a mostly enclosed, partially recessed room without mechanical ventilation and installed the liquid nitrogen freezer in an area particularly susceptible to an oxygen-deficient atmosphere.

32. FFG had no system, plan, or program to train and verify the competency of its employees when operating the liquid nitrogen freezer and working with or near hazardous liquid nitrogen. As a result, FFG employees at all levels of the organization were not aware of the hazards of liquid nitrogen and were unaware of precautions that should have been taken.

33. Had the liquid nitrogen freezer equipment and the entrance to the Line 4 freezer room been affixed with proper warning signage and labels, workers could have been made aware of the asphyxiation risks within the room.

34. FFG lacked an effective process safety management system to identify, evaluate, and control the hazards of the liquid nitrogen process. The lack of safety oversight by FFG leadership, absence of a systematic process hazard analysis, lack of written procedures, lack of any management of change practices, and failure to communicate hazards resulted in the unmitigated handling of a cryogenic asphyxiant by untrained and unprepared personnel. This incident could have been prevented had FFG practiced robust process safety management.

35. There is no regulation requiring employers handling or processing cryogenic liquid asphyxiants, such as liquid nitrogen, to adhere to industry guidance concerning proper design and safe handling, nor to implement a robust and systematic approach to process safety. Because it was not required to, FFG did not implement important process safety practices that could have either prevented the accidental release or reduced its severity.

36. There is no specific guidance from OSHA on the process safety practices necessary for the safe use of cryogenic asphyxiants.

Product Stewardship

37. Messer did not practice effective product stewardship prior to the incident. Messer was aware of several instances of FFG's process safety deficiencies and poor safety practices yet still commissioned the freezer to FFG. Using more effective product stewardship, Messer could have prevented this incident by

1) objecting to the placement of the freezer in a room particularly susceptible to oxygen deficiency, 2) refusing to commission its freezer equipment until FFG provided adequate atmospheric monitoring and alarm systems, 3) suspending supply of liquid nitrogen until safety deficiencies had been addressed, or 4) ending the relationship with FFG and removing its equipment.

38. Messer's newly developed customer safety inspection program provides a robust review of liquid nitrogen customer safety practices and can help prevent future incidents similar to the one at FFG.

5.2 CAUSE

The CSB determined the cause of the liquid nitrogen release was the failure of the immersion freezer's liquid level control system to accurately measure and control the liquid nitrogen level inside the freezer, which resulted from deformation of the system's bubbler tube component.

Contributing to the incident were 1) Messer's design of the freezer, which allowed the failure of a single level measurement device to result in an uncontrolled loss of containment of liquid nitrogen, 2) FFG's lack of any process safety management systems or practices that could have prevented the incident, 3) a lack of regulatory coverage for liquid nitrogen, which enabled FFG to elect not to implement process safety practices that could have prevented the incident, and 4) Messer's inadequate product stewardship practices, which resulted in Messer continuing to supply FFG with liquid nitrogen despite FFG's unsafe practices.

Contributing to the severity of the incident were 1) FFG's inadequate emergency preparedness, which resulted in at least 14 employees responding to the release by entering the freezer room or the surrounding area to investigate the incident or to attempt to rescue their coworkers, and 2) the absence of atmospheric monitoring and alarm devices that could have alerted workers to the presence of a hazardous atmosphere and warned them against entering.

6 RECOMMENDATIONS

To prevent future chemical incidents, and in the interest of driving chemical safety excellence to protect communities, workers, and the environment, the CSB makes the following safety recommendations:

6.1 GOLD CREEK FOODS[a]

2021-03-I-GA-R1

Include in the emergency action program provisions for proactively interacting with and informing local emergency response resources of all emergencies at the former FFG Plant 4 facility to which Gold Creek expects them to respond. At a minimum, Gold Creek should:

a) inform local emergency responders of the existence, nature, and location of hazardous substances at its facilities, including liquid nitrogen;

b) inform local emergency responders of the location of emergency-critical equipment such as bulk storage tanks, points of use, isolation valves, E-stop switches, and any other emergency equipment or systems with which emergency responders may need to interact; and,

c) provide local emergency responders with information, such as facility plot plans, engineering drawings, or other information needed to mount an effective emergency response.

6.2 MESSER LLC[b]

2021-03-I-GA-R2

Update the company product stewardship policy to:

a) include participation by Messer in customers' process hazard analyses (PHAs). The policy should require that these PHAs be conducted in a manner which conforms with CCPS *Guidelines for Hazard Evaluation Procedures* prior to the startup of a cryogenic freezing process;

b) require verification that proper signage, in accordance with CGA P-76 *Hazards of Oxygen-Deficient Atmospheres*, is displayed on and/or near equipment; and,

[a] Gold Creek has developed emergency response procedures that could have reduced the severity of this incident. Consequently, the CSB makes no recommendations to Gold Creek pertaining to the development of emergency preparedness policies or employee training for its emergency action program. Additionally, as of this report's publication, there are no liquid nitrogen freezing processes at the former FFG Plant 4 building which Gold Creek now operates. Consequently, the CSB makes no recommendation to Gold Creek related to process safety management practices for liquid nitrogen processes.

[b] After the incident Messer revised its freezer design to include multiple layers of protection against liquid nitrogen overflow. Messer also revised its Quality Control process and procedures to require verification of the presence of the necessary bubbler tube clamps and reordered the sequence of inspection steps to facilitate this verification. Consequently, the CSB makes no recommendation to Messer regarding the design of its immersion freezers or its Quality Control process.

c) require a facility and/or equipment siting review to ensure that emergency shutoff devices, including E-stops, are located such that they can be safely actuated during a release of liquid nitrogen.

2021-03-I-GA-R3

Create an informational product that provides Messer customers with information on the safety issues described in this report. In this informational product, recommend that Messer customers develop and implement effective safety management systems to control asphyxiation hazards from inert gases based on the guidance published in CGA P-86 *Guideline for Process Safety Management*, CGA P-12 *Guideline for Safe Handling of Cryogenic and Refrigerated Liquids*, CGA P-18 *Standard for Bulk Inert Gas Systems*, and CGA P-76 *Hazards of Oxygen-Deficient Atmospheres*.

6.3 U.S. Occupational Safety and Health Administration (OSHA)

2021-03-I-GA-R4

Update the Region 4 Poultry Processing Facilities Regional Emphasis Program to explicitly cover liquid nitrogen freezing processes. At a minimum, the update should encourage practices applicable to managing the hazards of using liquid nitrogen and other cryogenic asphyxiants, including process safety management practices, atmospheric monitoring, employee training and hazard awareness, and emergency preparedness and response.

2021-03-I-GA-R5

Update the Region 5 Regional Emphasis Program for Food Manufacturing Industry to explicitly cover liquid nitrogen freezing processes. At a minimum, the update should encourage practices applicable to managing the hazards of using liquid nitrogen and other cryogenic asphyxiants, including process safety management practices, atmospheric monitoring, employee training and hazard awareness, and emergency preparedness and response.

2021-03-I-GA-R6

Update the Region 6 Poultry Processing Facilities Regional Emphasis Program to explicitly cover liquid nitrogen freezing processes. At a minimum, the update should encourage practices applicable to managing the hazards of using liquid nitrogen and other cryogenic asphyxiants, including process safety management practices, atmospheric monitoring, employee training and hazard awareness, and emergency preparedness and response.

2021-03-I-GA-R7

Promulgate a standard specific to cryogenic asphyxiants. The purpose of this standard shall be the prevention and/or mitigation of hazards arising from the storage, use, and/or handling of these substances. The new standard shall reference applicable national consensus standards such as those published by the Compressed Gas Association and others, as appropriate. At a minimum the new standard shall:

a) Address requirements for the design, construction, and installation of process equipment storing or using cryogenic asphyxiants;

b) Require atmospheric monitoring where equipment storing or using cryogenic asphyxiants is located indoors;

c) Require emergency shutdown systems such that equipment storing or using cryogenic asphyxiants may be isolated during a release without endangerment;

d) Address requirements for employee training and hazard awareness specific to cryogenic asphyxiants;

e) Require an emergency action plan in accordance with 29 CFR 1910.38; and,

f) Address requirements for the use of process safety management elements such as process hazard analysis, management of change, procedures, and others deemed necessary through the rulemaking process to prevent and/or mitigate these hazards.

2021-03-I-GA-R8

Develop and publish a Guidance Document (similar to OSHA 3912-03 Process Safety Management for Explosives and Pyrotechnics Manufacturing) for process safety management practices applicable to processes handling compressed gases and cryogenic asphyxiants, including (at a minimum) the practices highlighted in this report.

6.4 Compressed Gas Association (CGA)

2021-03-I-GA-R9

Develop a comprehensive standard for the safe storage, handling, and use of liquid nitrogen in stationary applications, comparable to the guidance presented in CGA G-6.5 *Standard for Small Stationary Insulated Carbon Dioxide Systems.* At a minimum, the standard should include:

a) requirements for and guidance on the location, the maintenance, and the functional testing of atmospheric monitoring devices;

b) requirements for visible and audible alarm indication distinct from the building's fire alarm system and at a continuously attended location;

c) guidance on the design, function, periodic maintenance and testing, and location of room and emergency ventilation systems; and,

d) requirements for and guidance on the location of emergency shutdown devices including E-stops.

2021-03-I-GA-R10

Update P-76 *Hazards of Oxygen-Deficient Atmospheres*. At a minimum, the updated standard should:

a) require that atmospheric monitoring systems *shall be* utilized with processes, equipment, and piping systems capable of producing oxygen-deficient atmospheres;

b) require that atmospheric monitoring systems provide both visible and audible alarm indication distinct from a building's fire alarm system and at a continuously attended location;

c) require that processes, equipment, and piping systems capable of producing oxygen-deficient atmospheres *shall be* equipped with remotely operated emergency isolation valves (ROEIVs); and

d) include guidance on the adequate safe location of emergency stop devices. At a minimum this guidance should be harmonized with the requirements of ISO 13850 *Safety of machinery – Emergency stop function – Principles for design.* As necessary, augment the general guidance of ISO 13850 with guidance specific to processes, equipment, and piping using cryogenic asphyxiants and inert gases.

6.5 National Fire Protection Association (NFPA)

2021-03-I-GA-R11

Update NFPA 55 *Compressed Gases and Cryogenic Fluids Code* to:

a) require the use of atmospheric monitoring with cryogenic asphyxiants in accordance with industry guidance such as is contained in CGA P-76 *Hazards of Oxygen-Deficient Atmospheres* and CGA P-12 *Safe Handling of Cryogenic Liquids* in addition to CGA P-18 *Standard for Bulk Inert Gas Systems*; and,

b) include guidance on the adequate safe location of manual shutoff valves and devices such as emergency push buttons used to activate remotely operated emergency isolation valves (ROEIVs). At a minimum this guidance should be harmonized with the requirements of ISO 13850 *Safety of machinery – Emergency stop function – Principles for design.*

6.6 International Code Council (ICC)

2021-03-I-GA-R12

Update the International Fire Code to:

a) require the use of atmospheric monitoring with cryogenic asphyxiants in accordance with industry guidance such as is contained in CGA P-76 *Hazards of Oxygen-Deficient Atmospheres* and CGA P-12 *Safe Handling of Cryogenic Liquids* in addition to CGA P-18 *Standard for Bulk Inert Gas Systems*; and,

b) include guidance on the adequate safe location of manual shutoff valves and devices such as emergency push buttons used to activate remotely operated emergency isolation valves (ROEIVs) in cryogenic fluid service. At a minimum this guidance should be harmonized with the requirements of ISO 13850 *Safety of machinery – Emergency stop function – Principles for design.*